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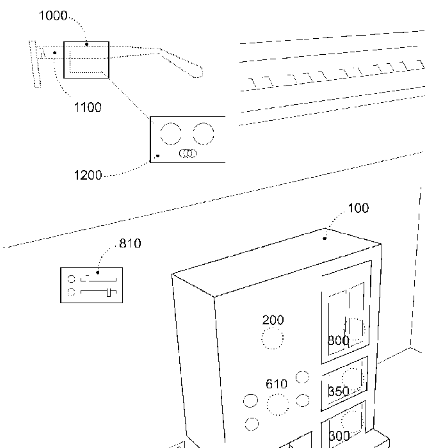
(54) Title: AUXILIARY PEDAL SYSTEM

FIG. 1

bFaaaP OVERVIEW



(57) Abstract: The purpose of the present invention is to provide an auxiliary pedal system (bFaaaP) comprising: a detector configured to detect a motion of a user to generate a detection signal; a processor configured to process the detection signal to generate an actuator-controlling signal; and an actuator configured to control a pedal in accordance with the actuator-controlling signal, wherein the motion involves a head movement; the head movement has an offset range within which the actuator is not actuated; and the actuator-controll ing signal is changeable in accordance with the user's preference. The pedal may be a piano pedal and the detector may be attached to eyeglasses. Also provided is a device controller having the above features. The present invention has a universal design for persons including kids and the challenged with leg disabilities and is applicable to any musical instruments with a pedal and any electric devices such as electric pianos and gaming controllers.

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Description

Title of Invention: AUXILIARY PEDAL SYSTEM

Technical Field

[0001] The present invention relates to an auxiliary pedal system and a device controller (gaming controller), and more specifically to a portable auxiliary piano pedal system in which an angle sensor is attached to eyeglasses; and a head tilt angle offset range and a multiplier selected by each user are used to execute an efficient way of controlling a device while providing a universal design for piano players including kids and persons with leg disabilities.

Background Art

[0002] The piano is a musical instrument having a keyboard, which is a row of keys depressed by the fingers of both hands of a player, and pedals, which are levers operated by feet of the player. The pedals include a soft pedal, a (optional) sostenuto pedal, and a sustain pedal (or a damper pedal or a sustaining pedal) (hereinafter, referred to as a "sustain pedal"). The sustain pedal is positioned to the right of the other pedals and is used more frequently than the other pedals. The sustain pedal raises all the dampers off the strings so that they keep vibrating after the player releases the key. Use of the pedal allows for a rich tonal quality so as to enhance piano performance. However, those whose legs cannot operate the pedals need auxiliary pedals or other assist devices. There are many types of auxiliary pedals for kids, but only a few commercially available auxiliary piano pedal devices for persons with disabilities (hereinafter, sometimes referred to as the challenged).

[0003] Easy-to-use piano pedal devices for persons with leg disabilities have long been sought. So far, there are several actual examples. In Europe, Dr. Rudiger Rupp and colleagues developed a wireless piano pedal controlled by the mouth (e.g., a tongue, bite) (Non Patent Literature 1). Another pedal assist device for a disabled pianist has also been known (Non Patent Literature 2). In U.S., Patent Literature 1 discloses a piano pedal operating device for people with disabilities, which device is operated by pressure exerted by the back of a user. In Japan, a pneumatic-control piano pedal system was designed for a professional pianist and Patent Literature 2 discloses a pedal actuator in this system.

[0004] Although not commercially available, a piano pedal performance support device has been proposed in Patent Literature 3 by YAMAHA Corporation. In this device, a continuous variate by an upper half of the body of a player with trouble in a lower half of the body is used to realize rich performance. The technical features involve a configuration where "a half point in a half pedal region of the pedal of the piano is predetermined and in the control table, an intermediate value between a minimum value and a maximum value of the detection signal corresponds to the predetermined half point".

[0005] The CanAssist team at University of Victoria created a head-activated piano pedal for a pianist (Non Patent Literature 3). The website recites "The CanAssist team set to work, creating a two-part technology: a mechanical device that sits on the floor and attaches to a piano pedal, and a headband containing a wireless sensor that measures changes in its own position. The sensor wirelessly communicates its position to the device on the floor, activating it to push down or release the pedal. So, wearing the headband, when Emily tilts her head down, the pedal is pushed down and the notes she is playing are sustained. By tilting her head up, the pedal is released".

Citation List

Patent Literature

[0006] Patent Literature 1: US Patent No. 9792885

Patent Literature 2: Japanese Patent No. 5367493

Patent Literature 3: Japanese Patent No. 4742574

Non Patent Literature

[0007] Non Patent Literature 1:

https://www.heidelberg-university-hospital.com/about-us/press-media/press-releases/singlenews/detail/News/despite-paraplegia-become-a-pianist/

Non Patent Literature 2:

http://www.pianoman.nl/pedal-assist-device-for-the-disabled-pianist.html

Non Patent Literature 3:

https://www.canassist.ca/EN/main/programs/technologies-and-devices/test-1/piano-arts.html

Summary of Invention

Technical Problem

[0008] In the previous actual piano pedal devices, oral motions such as tong movements or breathes have been used to actuate a pedal device. However, use of the oral motions is accompanied by sanitary problems and it is difficult for players to sing a song while playing the piano. In Japanese Patent No. 4742574, an upper half of the body of a player with leg disabilities may be used for the pedal device and examples of a motion to control a pedal include a head movement. The above head-activated piano pedal actually used a head movement detected by a wireless sensor placed on an upper front portion of the head by using a headband. The head movement such as a rotation rate or a tilt angle may be detected by, for instance, an accelerometer or gyroscope to give either a binary signal or a continuous signal. However, an effective way of detecting an actual specific motion to control the pedal device is neither defined nor put into practice. Thus, the efficient way of detecting and processing the motion of a player with leg disabilities should be sought further.

[0009] While Japanese Patent No. 4742574 defines the above half point in a half pedal region, the half point can be arbitrarily set between the minimum and the maximum of the detection signal and what the half point functionally means is indefinite. Thus, there is room for improvement in setting, if necessary, a position that causes the pedal to respond to a pedal-operating signal more effectively and quickly. In addition, how to set the position in the pedal device can be improved.

[0010] Besides, each pedal actuator seems to be operated either in a binary manner or in a continuous manner. However, recent progress in stepper motors has allowed for more fine control in a stepwise manner. Use of this control in the pedal device should also be sought.

[0011] The purpose of the present invention is to provide an auxiliary pedal system in which a non-oral head movement is utilized in an efficient manner. Another purpose of the present invention is to provide an auxiliary pedal system including a controller configured to process a head movement detection signal to generate and transmit an electric device-controlling command signal that replaces a pedal-mediated command signal. Also, provided is a device controller, such as a gaming controller, that uses a head motion to control a device.

Solution to Problem

[0012] A first aspect of the present invention provides an auxiliary pedal system comprising: a detector configured to detect a motion of a user to generate a detection signal; a processor configured to process the detection signal to generate an actuator-controlling signal; and an actuator configured to control a pedal in accordance with the actuator-controlling signal, wherein the motion involves a head movement; the head movement has an offset range within which the actuator is not actuated; and the actuator-controlling signal is changeable in accordance with the user's preference.

[0013] In the first aspect, the motion involves a head movement, in particular, a head movement of each person with leg disabilities and each kid. The head movement may involve a head rotation around each of the three axes of a X-Y-Z-axis coordinate system in which the X axis is in a horizontal direction extending from the user side to the piano side; the Z axis is in a another horizontal direction extending from the left side to the right side of the user; and the Y axis is in a vertical direction perpendicular to the X and Z axes. The Z-axis rotation, which reflects a nodding action of the user, is preferable, but the Z-axis rotation and the Y-axis rotation are also allowed. Preferably, a tilt angle with respect to the Z rotation axis is used, but a tilt angle with respect to the X rotation axis or the Y rotation axis may also be used. Each rotation rate may be detected by a gyroscope. An acceleration in each axis direction may also be used to detect the head movement and may be measured by an accelerometer. The tilt angle may be determined by the method described herein.

[0014] In the first aspect, the head movement has an offset range within which the actuator is not actuated. During piano performance, each player moves the head voluntarily, so that an unwanted head movement in terms of the detector's point of view may be mixed in a detection signal of the detector. To minimize the unwanted head movement, an offset range within which the actuator is not actuated is provided. Examples of the offset range may include offset ranges of a head tilt angle, a rotation rate, and an acceleration with respect to each of the X-, Y-, and Z-axes.

[0015] In the first aspect, the actuator-controlling signal is changeable in accordance with the user's preference. To realize an efficient way of controlling the actuator, the actuator-controlling signal may be generated in accordance with the user's preference. Because the user can select how fast the actuator responds to his/her head movement, the optimized response for each individual player can be achieved. The user's choice of the response rate is combined with the above offset range to give an efficient way of generating the actuator-controlling signal.

[0016] The pedal may be a piano pedal. As used herein, examples of the pedal include pedals for any devices such as musical instruments and vehicles such as automobiles. Examples of the piano include grand pianos, upright pianos, electrical pianos, organs, keyboards, and electones.

[0017] The actuator may comprise a stepper motor. Any kinds of actuator that can control and actuate a pedal may be used. However, a stepper motor is preferable in view of precise pedal control. Use of the stepper motor enables a pedaling speed and a duration at each pedal position to be controlled accurately. Also, a linear stepper motor is more preferable in order to move the motor reciprocally in a vertical direction while the pedal moves vertically.

[0018] The detector may be attached to eyeglasses, in particular, a side frame of the eyeglasses. To detect a tilt angle of the head of each user, the detector is preferably set horizontally. To achieve this objective, the detector may be attached along an eyeglasses side frame part between an eye and an ear of the user. Also, a case for the detector may have a slit or groove fit for the side frame. This makes it possible to arrange the detector substantially horizontally, so that the initial tilt angle is almost 0 degrees and it is easy for the user to set a head tilt angle offset value. The detector case may have a slit or groove fit for the eyeglasses side frame part so as to facilitate positioning of the detector. In addition, the detector may be a lightweight detector. This may facilitate the user's operation such as the player's piano performance because the user can control the device without feeling a burden on the head. Further, the detector may be fixed or detachable on the eyeglasses side frame part. If the user uses eyeglasses, the detector may be attached to his/her own eyeglasses. If the user does not use eyeglasses regularly, fake eyeglasses may be used for this purpose. Besides, how the detector looks in front of audience may be better when the detector is attached to a side frame of eyeglasses than when the detector is attached to a player's head by using a headband. A good appearance of a player wearing the detector is an important factor for piano performance.

[0019] The head movement may involve a head tilt angle and un upper limit of an offset range of the head tilt angle may be defined by the user. As used herein, the offset range involves, for instance, a regular nodding action of the head of a user. The upper limit of the offset range is preferably 10 degrees or less, more preferably 7.5 degrees or less, still more preferably 5 degrees or less, and still more preferably 3 degrees or less because Example 6 demonstrated that the above upper limit was actually selected by the study subjects and was found effective in suspending a tone by using each bFaaaP from the viewpoint of a quick response. That is, in view of a faster actuator response and persons with a neck disability, the less the offset value and the faster the actuator responds, the better. Particularly, 3 degrees were selected by a subject with leg disabilities and tracheotomy and were found effective. In view of cancelling the voluntary head movement, the upper limit of the offset range of the head tilt angle is preferably 3 degrees (2.5 degrees) or less, more preferably 5 degrees or less, still more preferably 7.5 degrees or less, and still more preferably 10 degrees or less, which were verified in the actual experiments of Example 9 as described below. Collectively, it has been experimentally demonstrated that the upper limit of the offset range (offset value) can range from 3 degrees to 10 degrees. Preferably, as described below, the head tilt angle in combination with the multiplier can be used to control a pedal action in an efficient manner.

[0020] The actuator-controlling signal may be modified by multiplying, by a multiplier, a value obtained by subtracting an upper limit of the offset range from the head tilt angle; and the multiplier is selected by the user and from 10 to 50, so that 2 to 10 degrees of the head tilt angle from the upper limit of the offset range are used to fully depress the pedal by the actuator. The multiplier may range from 1 to 50. When the multiplier is 10, 10 degrees of the head tilt angle from the upper limit of the offset range are used to fully depress the pedal by the actuator; when the multiplier is 20, 5 degrees of the head tilt angle are used; when the multiplier is 30, 3.3 degrees of the head tilt angle are used; when the multiplier is 40, 2.5 degrees of the head tilt angle are used; and when the multiplier is 50, 2 degrees of the head tilt angle are used. The multiplier can be selected according to the user's preference because a larger multiplier causes the pedal to be depressed quicker by the actuator and because a smaller multiplier cause the pedal to be controlled more precisely. Particularly, 20, 30, or 40 were preferably selected by the APEE study subjects as described in Example 6.

[0021] A free pedal play may be offset such that the pedal responds to the actuator-controlling signal and moves without a delay. Usually, the pedal has a free pedal play, so that in the case of the piano, for instance, the pedal may not lift dumpers initially. After the free pedal play is offset, the pedal starts functioning in this case. In view of the above, by offsetting the free pedal play at the initial state, the pedal responds to the actuator-controlling signal and moves without a delay. This should improve the actuator action in response to the actuator-controlling signal. Preferably, the free pedal play is determined manually but set by actually controlling the actuator by using an actuator controller.

[0022] The auxiliary pedal system may further comprise a portable housing having the actuator and a detachable weight(s), wherein the detachable weight resists reaction force of the actuator so as not to move the housing; and the housing has a sound-proof chamber including the actuator. Because the housing is portable, the system can be deployed almost anywhere. Also, because the weight is detachable, portability of the housing is improved when the weight is removed from the housing and is carried separately during transportation. In the case of the piano, the system may be fit for not only grand pianos in concert halls but also upright pianos and electric pianos at home. The detachable weight is provided so as to resist reaction force of the actuator. In this way, while the housing is portable, the housing is not easily moved during operation of the actuator. How heavy the weight is may be selected depending on the reaction force of a pedal of interest. For some pianos, two 5-kg weights are sufficient for this purpose. Another feature of the housing involves a sound-proof chamber. When the actuator is actuated in response to the detector signal, the actuator-associated sound (noise) is generated. To minimize or cancel the noise, this housing has a sound-proof chamber for the actuator. The chamber may have a sound absorber on walls. In the case of pianos, this can minimize the noise and increase tonal qualities of piano performance. Collectively, this configuration of the housing can simultaneously increase the portability, the stability, and the noise-cancelling effect of the housing.

[0023] A second aspect of the present invention provides a method for operating a pedal, comprising the steps of: detecting a motion of a user to generate a detection signal; processing the detection signal to generate an actuator-controlling signal; and controlling a pedal in accordance with the actuator-controlling signal, wherein the motion involves a head movement; the head movement has an offset range within which the actuator is not actuated; and the actuator-controlling signal is changeable in accordance with the user's preference. As described above, in the present method, the head movement is utilized in an efficient manner so as to control the pedal.

[0024] In the second aspect, the above described features may also be employed so as to enhance the pedal control.

[0025] A third aspect of the present invention provides an auxiliary pedal system comprising: a detector configured to detect a motion of a user to generate a detection signal; and a controller configured to process the detection signal to generate and transmit an electric device-controlling command signal that replaces a pedal-mediated command signal, wherein the motion involves a head movement; the head movement has an offset range within which the electric device does not respond to the head movement; and the electric device-controlling command signal is changeable in accordance with the user's preference.

[0026] In the third aspect, the controller configured to process the detection signal to generate and transmit an electric device-controlling command signal that replaces a pedal-mediated command signal. The third aspect does not have an actuator configured to actually control a pedal. Instead, in the third aspect, the electric device-controlling command signal replaces a command signal generated by a pedal of the electric device and is input directly into an electric device controller for performing a function of interest. In the third aspect, the controller may include a detector-side control unit and an electric device-side control unit. The detector-side control unit alone or with the electric device-side control unit may process the detector signal in accordance with the features of the present invention. The controller according to the third aspect preferably includes this detector-side controller and may transmit the electric device-controlling command signal in a wired or wireless manner. This extends the present invention to a variety of electric devices with or even without any pedal.

[0027] In the third aspect, the above-described features in the first aspect may be applicable.

[0028] The electric device may be an electric piano. Because each electric piano has an electric piano controller, the present invention may have a greater value for the electric piano than mechanical pianos. Just sending the electric device-controlling command signal to the electronic piano controller allows for an efficient way of enhancing piano performance by, in particular, the challenged with leg disabilities and kids.

[0029] A fourth aspect of the present invention provides a method for operating an electric device, comprising the steps of: detecting a motion of a user to generate a detection signal; and processing the detection signal to generate and transmit an electric device-controlling command signal that replaces a pedal-mediated command signal, wherein the motion involves a head movement; the head movement has an offset range within which the electric device does not respond to the head movement; and the electric device-controlling command signal is changeable in accordance with the user's preference.

[0030] In the fourth aspect of the present invention, the above-described features may be applicable so as to provide an efficient and universal way of controlling an electric device by almost everyone including the challenged with leg disabilities and kids.

[0031] A fifth aspect of the present invention provides a device controller comprising: a detector configured to detect a motion of a user to generate a detection signal; and a controller configured to process the detection signal to generate and transmit a device-controlling command signal, wherein the motion involves a head movement; the head movement has an offset range within which the device does not respond to the head movement; the device-controlling command signal is changeable in accordance with the user's preference; and the detector may be attached to eyeglasses.

[0032] In the fifth aspect of the present invention, the above-described features may be applicable so as to provide an efficient and universal way of fully operating a device by almost everyone including the challenged with leg disabilities and kids. In this aspect, the device controller may be a gaming controller. Use of a head motion may provide another way for inputting a user's action. Because this device controller is controlled by two elements involving an offset range and a multiplier changeable in accordance with the user's preference. This way may provide another way of controlling a game play.

Advantageous Effects of Invention

[0033] The present invention provides an auxiliary pedal system for any of users including persons with leg disabilities and kids. For the persons with leg disabilities, it is almost impossible to use a pedal, in particular, a piano pedal. However, this system makes it possible to use the pedal by almost everyone. In addition, a non-oral head movement is utilized in an efficient manner by using an offset value range of the head movement and by making each user control how fast an actuator or electric device responds to the head movement. This allows for efficient actuator control and electric device control.

Brief Description of Drawings

[0034] [fig.1]FIG. 1 is an overview of a bFaaaP according to an embodiment of the present invention.

[fig.2]FIG. 2 is a diagram illustrating a detector-side configuration and an actuator-side configuration of the bFaaaP.

[fig.3]FIG. 3A is a side view of an eyeglasses-attached detector according to an embodiment of the present invention. FIGS. 3B and 3C are other detectors according to embodiments of the present invention.

[fig.4]FIG. 4 is a detector-side processing flowchart I.

[fig.5]FIG. 5 is a detector-side processing flowchart II.

[fig.6]FIG. 6 is an actuator-side processing flowchart I.

[fig.7]FIG. 6 is an actuator-side processing flowchart II.

[fig.8]FIG. 8 is a MOV function flowchart.

[fig.9]FIG. 9 is a front view of a bFaaaP housing according to an embodiment of the present invention.

[fig.10]FIG. 10 is a rear view of the bFaaaP housing.

[fig.11]FIG. 11 is left-side (A) and right-side (B) cross-sectional views of the bFaaaP housing.

[fig.12]FIG. 12 is left-side (A) and right-side (B) views of the bFaaaP housing.

[fig.13]FIG. 13 is a top view (A) and a bottom view (B) of the bFaaaP housing.

[fig.14]FIG. 14 is intermediate cross-sectional views I (A) and II (B) of the bFaaaP housing.

[fig.15]FIG. 15 is enlarged views of a spacing plate member according to an embodiment of the present invention.

[fig.16]FIG. 16 is a front view of a bFaaaP4 housing with the size of each part indicated.

[fig.17]FIG. 17 is a rear view of the bFaaaP4 housing with the size of each part indicated.

[fig.18]FIG. 18 is left-side (A) and right-side (B) cross-sectional views of the bFaaaP4 housing with the size of each part indicated.

[fig.19]FIG. 19 is left-side (A) and right-side (B) views of the bFaaaP4 housing with the size of each part indicated.

[fig.20]FIG. 20 is a top view (A) and a bottom view (B) of the bFaaaP4 housing with the size of each part indicated.

[fig.21]FIG. 21 is intermediate cross-sectional views I (A) and II (B) of the bFaaaP4 housing with the size of each part indicated.

[fig.22]FIG. 22 is enlarged views of a spacing plate member of the bFaaaP4 with the size indicated.

[fig.23]FIG. 23 is notes of sheet music used in an APEE study.

[fig.24]FIG. 24 is a questionnaire provided to subjects in a protocol of the APEE study.

[fig.25]FIG. 25 is one of diagrams obtained in the APEE study and calculation formulas used for statistical analysis.

[fig.26]FIG. 26 is a graph showing head tilt angles of a subject during piano performance.

Description of Embodiments

[0035] Hereinafter, embodiments of the present invention are described in detail.

Terms and Definitions

[0036] The terms herein have usual meanings used by those skilled in the art. However, unless otherwise indicated, the following dictates the meaning of each term.

[0037] 1. The Piano

The piano is an acoustic, stringed musical instrument. As used herein, kinds of the piano include grand pianos, upright pianos, electric pianos (also, referred to as electronic pianos), keyboards, electones, and organs. Most modern pianos have a row of 88 keys consisting of 52 white keys and 36 shorter black keys. The pianos have usually two or three pedals.

As used herein, each specific side with respect to a piano relative to a player is defined as follows:

front side: the player's side relative to the piano;

rear side: an action mechanism side relative to a board behind a keyboard;

left side: the left side when viewed from the player;

right side: the right side when viewed from the player;

upper side: a higher position relative to a certain point; and

lower side: a lower position relative to a certain point.

In general, a musical tone is a steady periodic sound and is characterized by its duration, pitch, intensity (loudness), and timbre (or quality). When the duration of a tone is extended by pedal action, the tone is sustained. As used herein, the term "sustained tone" refers to this type of sustained tone.

In music, a note is the pitch and duration of a sound. The musical notes are usually represented by either C, D, E, F, G, A, and B or Do, Re, Mi, Fa, Sol, La, and Si. As used herein, the notes are represented by Do, Re, Mi, Fa, Sol, La, and Si.

[0038] 2. Piano Pedals

Piano pedals are foot-operated levers at the base of a piano. The pedals can change the piano's sound in various ways. The pedals usually include, from left to right, the soft pedal, the sostenuto pedal, and the sustain pedal. In some silent pianos, the middle sostenuto pedal may be replaced by a pedal with a muting function. The soft pedal produces a softer, more ethereal tone. The sustain pedal is used more frequently and raises all the dampers off the strings so that they keep vibrating after a player releases a key. The sustain pedal doubles as a pedal that allows the player to connect, into a legato texture, notes that otherwise could not thus be played.

When a piano pedal is depressed by a foot of a player, the player receives reaction force (kick back force) on the foot. As used herein, this force is referred to as "reaction force".

Each piano pedal usually has a free pedal play, which is also employed in clutch systems in automobiles.

As used herein, the term "free pedal play" refers to the distance or margin that a piano pedal can be depressed before the pedal begins to make an action such as sustaining a tone or softening a tone.

As used herein, the term “actuation width” refers to a distance from a point at a free pedal play offset value to a point where the pedal is depressed in accordance with a user’s operation of the present system.
As used herein, the term “actuation range value” refers to a distance from a point at a free pedal play offset value to a point where the pedal is fully depressed.

[0039] 3. Auxiliary Pedals

There are many kinds of auxiliary pedals for kids whose leg cannot reach a piano pedal when seated in front of the piano. Most of them are mechanical ones. Each auxiliary pedal usually assists a player to depress a piano pedal when the player depresses another pedal on such an auxiliary apparatus. However, persons with leg disabilities still cannot operate such auxiliary pedals.

Recently, universal design has been increasingly adopted for public equipment and facilities as well as buildings and housings. This wording “universal design” is used in the same context as of a “barrier-free” concept. However, there are very few commercially available auxiliary piano pedals that can be used for everyone including persons with leg disabilities. Embodiments of the present invention each provide an auxiliary piano pedal system that can be used for almost everyone including persons with leg disabilities. As used herein, this system is sometimes referred to as a “barrier-free assist as a pedal (bFaaaP)”.

[0040] 4. Head Movements

Head movements may include a head rotation around each of the three axes of a X-Y-Z-axis coordinate system in which the X axis is in a horizontal direction extending from the user side to the piano side (i.e., the above front side to the rear side); the Z axis is in another horizontal direction extending from the left side to the right side of the user (i.e., the above left side to the right side); and the Y axis is in a vertical direction perpendicular to the X and Z axes (i.e., the above lower side to the upper side). Any of the X-, Y-, and Z-axis rotations may be used. The rotations may involve a rotation rate and/or a head tilt angle. The Z-axis rotation reflects a nodding action of the user and may be used. A tilt angle with respect to the X rotation axis or the Y rotation axis may also be used. Also, an acceleration in each axis direction may be utilized for the present invention.

As used herein, a head tilt angle is determined as an angle (e.g., a pitch angle) with respect to a plane in parallel to the ground (i.e., a horizontal plane). As used herein, the “(head tilt angle) offset range” is a range within which an actuator is not actuated and may be set by each user. As used herein, the “(head tilt angle) offset value” is an upper limit of the offset range and is not 0 degrees.

As used herein, a “multiplier” is set by each user and a head tilt angle is multiplied by the multiplier. Each user can select the multiplier according to their preference.

[0041] 5. Subjects (Users or Piano Players)

As used herein, piano players (users) are grouped into Class I, II, and III subjects. Class I subjects are those who can depress a pedal by their own foot (sometimes herein referred to as adults). Class II subjects are those who can move their legs but cannot operate a piano pedal without a conventional auxiliary pedal (e.g., children (kids)). Class III subjects are those who have leg disabilities and cannot operate a piano pedal even with the conventional auxiliary pedal (e.g., persons with leg disabilities) (hereinafter, persons with leg disabilities are sometimes referred to as the challenged with leg disabilities). If Class II subjects also fall under Class III subjects (e.g., children with leg disabilities), these subjects are treated as Class III subjects. As used herein, professional pianists are also referred to as piano players.

[0042] 6. Auxiliary Pedal Effect Evaluation Study (APEE study)

Effects of this auxiliary pedal system may be evaluated by measuring how long each tone is sustained by actuating the actuator of this system. To provide a piece of evidence showing the effects, the above Class I to III subjects are tested. As an approach to measuring the sustained tone, a tone-vibrating area (hereinafter, sometimes referred to as a “TVA”) of each sound waveform is measured and evaluated with respect to each class and individual class subjects. The detailed protocol is described in Example 6 below. Hereinafter, this test is called an auxiliary pedal effect evaluation study (APEE study).

[0043] 7. Terms in Statistics

Standard terms for statistics are used herein, including a t-test, an F test, one-way ANOVA (analysis of variance), an average, a standard deviation (SD), and a p-value.

[0044] Detailed Description of Embodiments

Embodiment 1

(A) Overview of bFaaaP According to Embodiment of the Present Invention

A bFaaaP is an auxiliary pedal system according to an embodiment of the present invention and may include a detector, a processor (controller), and an actuator. FIGS. 1 and 2 show an overview of the bFaaaP according to this embodiment. The detector is configured to detect a motion of a user to generate a detection signal. Next, the processor is configured to process the detection signal to generate an actuator-controlling signal. Then, the actuator is configured to control a pedal in accordance with the actuator-controlling signal. Here, the motion involves a head movement; the head movement has an offset range within which the actuator is not actuated; and the actuator-controlling signal is changeable in accordance with the user’s preference. So, the bFaaaP may have a three-member configuration. The following describes, in detail, each member by referring to the Drawings.

[0045] (A-1) Detector (Angle Sensor)

(1) Gyroscopes
Each gyroscope is a device used for measuring or maintaining orientation and angular velocity. Different kinds of gyroscopes are present, but any of the gyroscopes can be used in the present invention as long as they can detect a head movement and give a detection signal.
(2) Accelerometers
Each accelerometer is a device that measures proper acceleration. There are different kinds of accelerometers and they are widely applicable to various fields. Any of the accelerometers can be used in the present invention as long as they can detect a head movement and give a detection signal.
(3) Geomagnetic Sensors
Each geomagnetic sensor is a device used to determine the orientation of objects with respect to the magnetic field surrounding the earth. The sensor may also be used to determine orientation with respect to a local magnetic field, a local attenuation of the earth's magnetic field, or used as a pointer to track moving magnetic potentials.
(4) Combinations
The present detector may include a combination of a gyroscope, an accelerometer, and/or a geomagnetic sensor. This detector may further include another device for detecting a head movement.
(5) Other Devices
The head movement may be monitored by a camera of a smartphone or a web camera and the camera image may then be sent to an image processor on the smartphone or an internet server for analysis. The head movement image may be processed though deep learning to generate a detector signal.
(6) How to Attach Detector to Head
FIG. 3 shows how to attach a detector (case) to the head of a user. In a preferable embodiment, the detector 1000 is attached to a side frame 1100 of eyeglasses. In FIG. 3, the size (mm) of each part is provided for illustration purpose, and the size is not limited to those numbers. FIG. 3A is a side view according to this embodiment. FIG. 3B shows the case where a detector case 1010 has a slit 1020 fit for the side frame 1100. FIG. 3C shows the case where a detector case 1010 has a groove 1020 fit for the side frame 1100. In each case, a sensor module 1050 is enclosed in the detector case 1010. Because the side frame 1100 is placed substantially horizontally between an ear and an eye of the user, the sensor module 1050 can also be set substantially horizontally. Here, the detector case 1010 may be attached on the left side or on the right side, whichever is preferred by each user. In addition, the detector or the side frame may be further fixed by using a rubber band (s) or a velcro tape(s).

[0046] <u>(A-2) Communication Devices (Transmitter and Receiver and Solid and Dashed</u>

Lines)

(1) Wireless Communication

Any kinds of wireless communication devices may be used as long as the devices allow for communication, for instance, between the detector and the processor (controller), between the processor (controller) and the actuator (an actuator controller or a driver for the actuator), and/or between the controller and the electric device (a controller for the electric device). Examples include Wi-Fi devices, Bluetooth devices, and BLE (Bluetooth Low Energy) devices.

(2) Wired Communication

Any kinds of cables (e.g., codes, wires, lines) may be used for the above communication.

Meanwhile, any of units of interest may be integrated or provided as a built-in unit.

[0047] (A-3-1) Processor or Controller (e.g., Data Processor and Actuator Controller)

(1) Arduino Hardware

Arduino hardware includes single-board microcontrollers and microcontroller kits for building digital devices and interactive objects that can sense and control objects in the physical and digital world. Arduino board designs use a variety of microprocessors and controllers. The boards are equipped with sets of digital and analog input/output (I/O) pins that may be interfaced to various expansion boards or breadboards (shields) and other circuits. The boards feature serial communications interfaces, including Universal Serial Bus (USB) on some models, which are also used for loading programs from personal computers. Examples of the Arduino hardware include Arduino Uno, Arduino Leonardo, Arduino Mega, Arduino Nano, and M5Stack.

The microcontrollers are typically programmed using a dialect of features from the programming languages C and C++. In addition to using traditional compiler toolchains, the Arduino project provides an integrated development environment (IDE) based on the Processing language project.

(2) Other Processors

Any kinds of processers that can process a detection signal from the detector to generate a control (command) signal may be adopted. Examples may include SPRESENSE series (SONY, Inc.). Each processor may also be placed on an IC board.

(3) How Each Processor Is Arranged

Each processor may comprise a detector-side control unit and an actuator-side control unit as shown in FIG. 2. The detector-side control unit alone or with the actuator-side controller may process the detector signal in accordance with the features of the present invention.

(4) Adjuster for Head Motion Offset Range and Adjuster for Multiplier

The processor (controller) may have adjusters (input units) (i.e., 1210 and 1220 in

FIG. 3A) for obtaining a head tilt angle offset range value and a multiplier as shown in FIG. 2. Any kind of the adjusters may be used. Examples of each adjuster include buttons, dials, levers, sliders, and switches. In addition, an on/off switch 1230 for the processor may be included.

(5) Adjuster for Free Pedal Play Offset Value and Adjuster for Actuation Range Value

The processor (controller) may also have additional adjusters (initializer 810 in FIG. 1) for obtaining a free pedal play offset value and an actuation range value as shown in FIG. 2. Any kind of the adjuster (initializer) may be used. Examples of the adjuster include buttons, dials, levers, sliders, and switches.

[0048] (A-3-2) Controller Programs

Controller programs may be written in any of known computer programming languages. Examples include Java, C, C++, C#, Python, JavaScript, PHP, Visual Basic, Ruby, Perl, Swift, Go, and Processing. When the controller is an Arduino-based controller, the Arduino IDE may be used to write and debug each program. The program may be installed in the Arduino board through a USB.

[0049] (A-4) Actuators (Motors)

(1) Stepper Motors

Each stepper motor or step motor or stepping motor (hereinafter, sometimes referred to as a stepper motor) is a DC electric motor that divides a full rotation into a number of equal steps. The motor position can then be commanded to move and hold at one of these steps. Any kinds of motors, however, may be used in the present invention as long as each motor can be used to depress a pedal. In view of precise control of pedal action, stepper motors are used preferably.

(2) Linear Motors

Each linear motor is an electric motor that has its stator and rotor "unrolled" so that instead of producing a torque (rotation) it produces a linear force along its length. For pedal action, which pedal reciprocally moves up and down, it may be advantageous to use a linear motor. For the present invention, it is preferable to use a linear stepper motor as the actuator.

[0050] (A-5) Power Source

Each power source may be an AC power source or a DC power source. The AC voltage is converted to DC voltage by an AC/DC converter to operate the actuator. In addition, each power source may be a battery. Examples of the battery include lithium ion secondary batteries. The number of the power sources or batteries is not limited. In FIG. 2, specific connections to and from each power source are omitted for description purpose.

[0051] (B) Specific Configuration of Auxiliary Pedal System (bFaaaP) According to This Embodiment

(B-1) Units in This System

A configuration of this embodiment is shown in FIG. 2 and includes an angle sensor (1), a data processor (2), a transmitter (3), a receiver (4), an actuator controller (5), an actuator (6), and detector-side and actuator-side power sources (7).

[0052] (B-2) Description of Each Unit

(B-2-1) Angle Sensor

The angle sensor (1) detect a change in tilt angle of the head of a user to generate a detector signal as an analog voltage or a digital value. Then, the detector signal is sent to the data processor (2).

[0053] The following points are considered for the angle sensor:

1) The angle sensor has a sufficient precision;

2) There is no burden imposed while a player moves the body;

3) The sensor is durable;

4) The sensor does not prevent a player from singing a song during performance;

5) The sensor should be small;

6) The sensor should be commercially available or easily constructed;

7) The sensor should be inexpensive.

[0054] The angle sensor may be attached to a head portion. In view of easily detecting a head motion, in particular, a tilt angle, the sensor may be attached to a side frame of eyeglasses. This is because the side frame is placed horizontally between an eye and an ear. This makes it easy to detect a head tilt angle. Also, the angle sensor is enclosed in a case with a slit or groove. The slit or groove is fit for the side frame so as to tightly fix the angle sensor in a proper position. The angle sensor may also be attached to a cap, a hat, a hair band, a hair accessory, or an ear accessory as long as the angle sensor can detect a head angle with precision.

[0055] The weight of the detector (angle sensor) may be between 0.1 g to 20 g. From the viewpoint of detecting a head motion without feeling a burden on the head and executing required performance, the weight is preferably from 0.1 g to 15 g, more preferably from 0.5 g to 10 g, and still more preferable from 1 g to 5 g. The size of the detector (angle sensor) may be 35 mm (horizontal length) x 30 mm (vertical length) x 15 mm (width) as shown in FIG. 3. However, the sizes are not limited to the above numbers and may be larger or smaller as long as the detector can execute proper performance while a user does not feel a burden on the head. Here, examples of the horizontal length include 1, 2, 3, 4, 5, 10, 15, 20, 25, 30, 35, 40, or 50 mm; examples of the vertical length include 1, 2, 3, 4, 5, 10, 15, 20, 25, 30, 35, 40, or 50 mm; and examples of the width include 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 15, 20, 25, or 30. Each value may be between any two of the above numbers. From the viewpoint of a lightweight and compact device, the size of the detector (angle sensor) is preferably about 35 mm (horizontal length) x about 30 mm (vertical length) x about 15 mm (width). Further, the detector may be attached to a frame part on the right side or the left side or frame parts on both the sides of eyeglasses.

[0056] Further, the detector (angle sensor) may be combined with an eyeglasses-type display (e.g., smart glasses) on which information such as sheet music or instructions can be displayed. This may make it easier to read sheet music while a piano player depresses a pedal by using a bFaaaP.

[0057] <u>(B-2-2) Data Processor</u>

The data obtained by the angle sensor (1) is subject to filter processing to reduce noise and drifts. A head tilt angle offset value and a user-defined multiplier are set by an offset adjuster and a multiplier adjuster, respectively, that are connected to the data processor (2). The offset value is subtracted form an actual angle and the resulting value is multiplied by the multiplier. If the value obtained is not within a suitable range (e.g., exceeds an upper limit), the value is corrected and is sent to the transmitter (3) as control data.

[0058] The detector-side controller including the data processor and the following transmitter may be selected in view of the following points:

1) The data processor can process data from the angle sensor without a delay;

2) The detector controller is small;

3) The detector controller consumes less energy;

4) The transmitter is enclosed in the controller;

5) The data processor has two or more A/D converters;

6) The data processor has an enough number of universal digital ports (for communication with the angle sensor);

7) The controller has an I2C communication port (for communication with the angle sensor);

8) The data processor can be easily debugged; and

9) The controller can be battery-operated.

[0059] <u>(B-2-3) Transmitter</u>

The transmitter (3) transmit the above control data via the receiver (4) to the actuator controller (5). The communication may be a wireless communication or a wired communication using an appropriate specification and a suitable protocol.

[0060] <u>(B-2-4) Receiver</u>

The receiver (4) receives the above control data by using the above communication protocol and sends the data to the actuator controller (5).

[0061] <u>(B-2-5) Actuator Controller</u>

The actuator controller (5) is initialized by determining an operation starting point (defined by a free pedal play offset value) and an actuation range value. Next, the incoming data (i.e., a value from 0 to an upper limit) is converted to fit for an actuation range. Then, the converted value is sent, as a command value, to the actuator (6).

[0062] The actuator-side controller including the actuator controller and the above receiver may be selected in view of the following points:

1) The actuator controller consumes less energy;

2) The receiver is enclosed in the controller;

3) The data processor has two or more A/D converters;

4) The data processor has an enough number of universal digital ports (for communication with the actuator);

5) The data processor can be easily debugged; and

6) The controller can be battery-operated.

[0063] <u>(B-2-6) Actuator</u>

The actuator (6) may include an actuator driver and a motor and is operated in accordance with the command value from the actuator controller.

[0064] The actuator may be selected in view of the following points:

1) The actuator can exert sufficient force to move a pedal;

2) The response speed of the actuator is sufficient so as not to cause a delay of performance of a player;

3) Noise of the actuator is acceptable in view of performance of a player; and

4) The actuator can be battery-operated.

[0065] <u>(B-2-7) Power Source</u>

Each power source may be an AC power source or a DC power source. The AC voltage is converted to DC voltage by an AC/DC converter to operate the actuator (6). In addition, each power source may be a battery. Examples of the battery include lithium ion secondary batteries.

[0066] <u>(C) Controller Programs</u>

The detector-side and actuator-side programs may be created such that a head tilt angle is detected and a pedal of a piano is depressed or released in according to a change in the angle. Here, the following points are considered.

[0067] (1) An attitude of each player is different among players, so that an initial head tilt angle and a voluntary head movement vary depending on the players. Thus, a head tilt angle offset value (i.e., an upper limit of an offset range) is introduced and an actuation starting point can be freely set by each player.

[0068] (2) How much the head is tilted by each player varies among players. Thus, a multiplier is introduced and how fast the actuator responds can be set and controlled by each player.

[0069] (3) Different pianos have different actuation stating points of a pedal. Thus, a free pedal play offset value is introduced and each actuation start point can be adjusted for each piano.

[0070] (4) Different pianos have different actuation range values. Thus, each actuation range value can be set and adjusted for each piano.

[0071] (5) A head tilt angle should be stably obtained as a detection signal. Thus, the detection signal may be subject to Madgwick filter processing to reduce noise and drifts.

[0072] (C-1) Programs for Detector-side Controller (Processor)

The flowcharts I and II of FIGS. 4 and 5 show how to detect and process a motion of a head to generate a detector signal according to an embodiment of the present invention. The numbers and specific operations may be changed depending on situations. A Nassi-Shneiderman diagram (NSD) is used to describe the processing. The "T" represents TRUE and indicates that a subject condition is met. The "F" represents FALSE and indicates that the subject condition is unmet.

[0073] At S100, global variables are defined and declared. From S101 to S105, ports, serial communication, an EEPROM, BLE, and an MPU9250 are set.

[0074] In a loop, local variables are defined at S106. From S107 to S109, sensor data (an accelerometer, a gyroscope, and a geomagnetic sensor) is acquired to calculate angle data (e.g., a pitch angle). Then, the resulting data is subjected to Madgwick filter processing at S110.

[0075] At S111, an offset value is subtracted from the pitch angle. At S112, whether the resulting pitch angle is less than 0 is determined. If true, the pitch angle is set to 0. If not, the process goes to S114.

[0076] At S114, the pitch angle is multiplied by a multiplier of from 1 to 50. At S115, whether the resulting pitch angle is larger than 99 is determined. If true, the pitch angle is set to 99. If not, the process goes to S117. The resulting range (i.e., from 0 to 99) is converted to an actuation width (i.e., from 0 to 47) to operate the actuator by the below-described actuator-side controller program.

[0077] Here, in order to fully depress a pedal by the actuator, the upper limit (i.e., 99) of the above range is used. That is, if the multiplier is 10, 10 degrees of pitch angle (head tilt angle) from the offset position is used to fully depress the pedal; if the multiplier is 20, 5 degrees of the pitch angle is used; if the multiplier is 30, 3.3 degrees of the pitch angle is used; if the multiplier is 40, 2.5 degrees of the pitch angle is used; and if the multiplier is 50, 2 degrees of the pitch angle is used.

[0078] At S117, the numerical pitch angle is converted to a string for BLE communication. At S118, the resulting string is transmitted through BLE communication.

[0079] Next, whether the offset adjuster is on is checked at S119. When the offset adjuster value is changed (S120), an offset volume value (i.e., an internal value of the adjuster; an analog volume value is converted to a digital value by an A/D converter; the same applies to the following) is read at S121; the value is converted to from -45 to 45 (tilt angle degrees) at S122; and this converted value is stored as a new offset value (an upper limit of an offset range of interest) at S123.

[0080] Then, whether the multiplier adjuster is on is checked at S124. When the multiplier adjuster value is changed (S125), a multiplier volume value (i.e., an internal value of the adjuster) is read at S126; the value is converted to from 1 to 50 at S127; and this converted value is stored as a new multiplier at S128.

[0081] Subsequently, the process goes back to S106 to repeat the loop.

[0082] (C-2) Programs for Actuator-side Controller (Processor)

The flowcharts I and II of FIGS. 6 and 7 show how to control an actuator.

[0083] At S200, global variables are defined and declared. A union is defined at S201. From S202 to S204, a BLE server, serial communication ports, and I/O ports are set.

[0084] At S205, the actuator is set at the original position.

[0085] At S206, the BLE communication is initialized and started.

[0086] If the BLE is off (S207), the process goes to S218. If the BLE is on, whether both the offset adjuster (i.e., the adjuster for a free pedal play offset value) and the actuation range value adjuster are off is checked at S208. If both the adjusters are off, the process goes to S209. Otherwise, the process goes to S218. If data is received (S209), whether this data is different from the previous data is determined at S210. If so, the transmitted string from the detector-side controller is converted to a number at S211. At S212, the string is stored as a new previous data. Then, the resulting number (0 to 99) is converted to an actuation width. If the converted actuation width is equal to or more than 47 (i.e., an upper limit of a command value for operating the actuator) (S214), the value is set to 47 at S215. After that, the resulting value is converted to bits by using the above union (S216) so as to operate the actuator by calling the actuation routine (the following MOV function) at S217.

[0087] Next, whether the free pedal play offset adjuster is on is checked at S218. If the offset adjuster is off, the process goes to S227. If the offset adjuster is on, the free pedal play offset volume value (i.e., an internal value of the adjuster) is read and divided by 44 at S219. Because the A/D converter of an ESP32 has 12-bit resolution (i.e., 2^{12}; an integer from 0 to 4096) and the upper limit of the actuation width is set to 47 in this case, the value of the A/D converter value is first divided by 47 to give a number of from 0 to about 88. Here, if the number is an odd number (S220), the process goes to S227. Next, this value (maximum 88) is divided by 2 at S221 to give a number less than 47 (i.e., an upper limit of a command value for operating the actuator). If the number is equal to or more than 47 (S222), the number (value) is set to 47 at S223. The same applies to the case of the actuation range value adjuster. After that, the resulting value is converted to bits by using the above union (S224) so as to operate the actuator by calling the actuation routine at S225. Finally, the value is stored as the free pedal play offset value at S226.

[0088] Next, whether the actuation range value adjuster is on is checked at S227. If the actuation range value adjuster is off, the following S228 to S239 are skipped and the loop is repeated. If the actuation range value adjuster is on, the actuation range volume value (i.e., an internal value of the adjuster) is read and divided by 44 at S228. If the resulting value is an odd number, the steps from S230 to S239 are skipped and the loop is repeated. Next, this value (maximum 88) is divided by 2 at S230 to give a number (actuation range value) less than 47. Then, the above free pedal play offset value is subtracted from the number at S231. If the resulting number is 0 or less (S232), this value is set to 0 at S233. After that, the free pedal play offset value is added to this value at S234. If the resulting value is 47 or more (S235), the value is set to 47 (the maximum of the actuation width) at S236. Thereafter, the resulting value is converted to bits by using the above union (S237) so as to operate the actuator by calling the actuation routine at S238. Subsequently, the value is stored as the actuation range value at S239. Then, the process goes back to S207 and the loop is repeated.

[0089] (C-3) MOV Function Processing

The flowchart of FIG. 8 shows how to operate an actuator by using the above actuation routine.

[0090] At S300, while the actuator is busy, the process does not go to the next step. If the actuator is available for setting, appropriate bits (Mo0 to Mo5) are set for the actuator operation (at steps from S301 to S306). Then, a 5-ms delay is placed at S307 and the START bit is turned on at S308 to actuate the actuator. At S309, while the actuator is busy, the process does not go to the next step. If not, the START bit is turned off at S310.

[0091] (D-1) How to Operate bFaaaP

First, a motion of a user is detected so as to generate a detection signal. Next, the detection signal is processed to generate an actuator-controlling signal. Then, a pedal is controlled by the actuator in accordance with the actuator-controlling signal. Here, the motion involves a head movement; the head movement has an offset range within which the actuator is not actuated; and the actuator-controlling signal is changeable in accordance with the user's preference. The head movement may involve a head tilt angle and the offset range of the head tilt angle may be defined by the user. Examples of the upper limit of the offset range include 0.5, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, or 20 degrees. The upper limit may be a value between any two of the above numbers. However, 3, 5, or 10 degrees were particularly selected by the APEE study subjects as described in Example 6. In addition, a basal value of the actuator-control signal may be multiplied by 1 to 50 set by the user. Examples of the multiplier include 1, 2, 3, 5, 10, 15, 20, 25, 30, 35, 40, 45, or 50. The multiplier may be a value between any two of the above numbers. However, 20, 30, or 40 was particularly selected by the APEE study subjects as described in Example 6.

[0092] Before actual pedal operation, a free pedal play may be offset such that the pedal responds to the actuator-controlling signal and moves without a delay. In this case, the height of the actuator may be programmatically controlled by setting an initial height of the actuator by using an input unit like the adjuster described above. The initial height may also be set manually in some cases.

[0093] (D-2) Operation Protocol

Steps

(1) Switch on the actuator-side controller by plugging in the first power supply.

(2) Switch on the detector-side controller by plugging in the second power supply.

(3) Use an adjuster for a free pedal play offset value to set the offset value while a key on a keyboard is depressed intermittently.

(4) Use an adjuster for an actuation range value to set the actuation range value such that the pedal is fully depressed.

(5) Use an adjuster for a head tilt angle offset value to set the offset value in accordance with each user's preference while the pedal is actuated by the bFaaaP.

(6) Use an adjuster for a multiplier to set the multiplier in accordance with each user's preference while the pedal is actuated by the bFaaaP.

(7) Operate the bFaaaP while each user, for example, plays the piano.

[0094] The order of steps (3) and (4) may be changed. Once steps (3) and (4) are completed for a specific piano, each user may omit these steps next time. The order of steps (5) and (6) may be changed but may be better to keep this order.

[0095] Embodiment 2

Piano Pedal

According to an embodiment of the present invention, the pedal is a piano pedal. Kinds of the piano include grand pianos, upright pianos, electric pianos, keyboards electones, and organs.

[0096] Embodiment 3

How to Attach Detector to User's Head

In order to detect a head movement of a user, the detector is preferably attached to a portion of the head of the user. Examples of a member (e.g., a tool, a device) for attaching the detector to the head include, but are not particularly limited to, caps, hats, headbands, ear attachments, and eyeglasses. According to a preferable embodiment of the present invention, the detector is attached to eyeglasses as shown in FIG. 3A. More specifically, the detector may be attached to an eyeglasses side frame part between an eye and an ear of the user. This makes it possible to arrange the detector substantially horizontally, so that the initial tilt angle is almost 0 degrees and it is easy for the user to set the tilt angle offset value. As shown in FIG. 3, a detector 1000 may have a slit (FIG. 3B) or a groove (FIG. 3C) fit for the eyeglasses side frame part so as to facilitate positioning of the detector. In addition, the detector may be a lightweight detector. This may facilitate the user's operation such as the player's piano performance because the user can control the device without feeling a burden on the head. Further, the detector may be fixed or detachable on the eyeglasses side frame part. If each user uses eyeglasses, the detector may be attached to his/her own eyeglasses. If each user does not use eyeglasses regularly, fake eyeglasses may be used for this purpose. Besides, how the detector looks in front of audience may be better when the detector is attached to a side frame of eyeglasses than when the detector is attached to a player's head by using a headband. A good appearance of a player wearing the detector is an important factor for piano performance.

[0097] Embodiment 4

(A) Configuration of Portable Housing

According to an embodiment of the present invention, each of the above auxiliary pedal systems further comprises a portable housing having the actuator and a detachable weight(s) wherein the detachable weight(s) resists reaction force of the actuator so as not to move the housing; and the housing has a sound-proof chamber including the actuator.

[0098] (1) How to Construct Portable Housing

FIGS. 9 to 15 show the structure of a housing 100 according to an embodiment of the present invention. FIG. 9 is a front view; FIG. 10 is a rear view; FIG. 11 is left-side and right-side cross-sectional views; FIG. 12 is left-side and right-side views; FIGS. 13 to 14 are cross-sectional views including a top view, a bottom view, and intermediate views; and FIG. 15 is enlarged views of a spacing plate member 630. The housing 100 has a sectioned structure and includes a sound-proof chamber 200, a first weight chamber 300, a second weight chamber 350, and an actuator controller chamber 800 as shown in FIGS. 9 and 10. The chambers are built by using plate members and regular fixing procedures (e.g., a bolt and a nut, a screw, an adhesive) known in the art. Each plate member may be made of any material such as wood, plastic, metal, and/or ceramic as long as the housing can be constructed so as to support devices (e.g., an actuator, a controller, a driver, an initializer, weights).

[0099] The sound-proof chamber 200 of FIGS. 9 to 12 and 14 is defined by a ceiling plate member 540, a second side plate member 560, a third side plate member 570, a rear plate member 580, a front plate member 590, and a sound-proof chamber bottom plate member 510. A sound absorber 210 is attached to all or part of the above plate members. A body portion 610 of an actuator (i.e., a linear stepper motor; an electric cylinder) 600 is fixed via an actuator attachment plate 615 to the front plate member 590 as shown in FIG. 11A and 14A. The actuator attachment plate 615 has 4 through-holes into which inner rubber plugs 592b1 to b2 and 594b1 to b2 and corresponding outer rubber plugs 592a1 to a1 and 594a1 to a2 are inserted while plug bolts are used to fix and fit each plug in each hole. These rubber plugs serve as sound absorbers as well as vibration-proof materials. The sound-proof chamber bottom plate member 510 has an elongated opening 250 as shown in FIG. 11A. A leg portion of the actuator 600 extends downwardly through the opening 250 from the body portion and moves reciprocally so as to depress and release a pedal. A spacing plate member 630 is attached to the bottom of the leg portion so as to avoid a direct contact between the actuator and the pedal. FIG. 15 is enlarged views of the spacing plate member 630. The spacing plate member may be made of any material as long as damage to the pedal can be avoided. Further, the spacing plate member 630 may be covered with a non-woven or woven fabric or a rubber material depicted as a dashed line in the bottom figure of FIG. 15. Collectively, this configuration of the sound-proof chamber 200 can be used to reduce operation noise generated during operation of the actuator 600.

[0100] The first weight chamber 300 of FIGS. 9 to 11 and 14B is defined by a first side plate member 550, the second side plate member 560, a first horizontal plate member 520, a floor plate member 500, and the rear plate member 580. The second weight chamber 350 of FIGS. 9 to 11 is defined by the first side plate member 550, the second side plate member 560, the first horizontal plate member 520, a second horizontal plate member 530, and the rear plate member 580. The front side of each chamber is exposed to the air as shown in FIG. 9. In each of the first and second weight chambers, a bolt (310 or 360) is fixed to and protruded from the first side plate member 550 and the tip pf each bolt does not reach the second side plate member 560 as shown in FIG. 9. Because of this configuration, a detachable weight(s) can be placed in each chamber. Weights 700a1-n each have a hole through which the bolt 310 is inserted and are placed in the first weight chamber 300. In addition, weights 750b1-n each have a hole through which the bolt 360 is inserted and are placed in the second weight chamber 350. The total weight of all the weights should be enough to resist pedal reaction force of the actuator during operation. The pedal reaction force was measured in Example 8. With reference to the values in Example 8, the total weight may be set to 8 to 10 kg. Use of the detachable weights can increase portability of the housing.

[0101] The actuator controller chamber 800 of FIGS. 9 to 11 and 14A is defined by the first side plate member 550, the second side plate member 560, the second horizontal plate member 530, the ceiling plate member 540, and the rear plate member 580. The front side of this chamber is also exposed to the air. This chamber may house an initializer 810 (adjusters for a free pedal play offset value and an actuation range value), an actuator controller 820, and a driver unit (e.g., a driver and a power supply). These devices may be fixed or detachable. The exposed front side of this chamber may be further provided with a plate member. This additional plate may improve an appearance of the housing because the devices cannot be seen from the outside.

[0102] A vibration-proof sheet 900 of FIGS. 9 to 11 may be attached to the bottom of the floor plate member 500 so as to reduce actuator-mediated vibrations and stabilize the housing. Any material may be used for this sheet as long as the above objectives are achieved.

[0103] The housing 100 may be placed next to a pedal as shown in FIG. 1. When a sustain pedal is depressed by the actuator 600, the housing may be placed on the right side of the pedal. When a soft pedal is depressed by the actuator 600, the housing may be placed on the left side. The pedal may be of a grand piano or an upright piano.

[0104] To further stabilize the housing 100 on the floor, an adhesive tape may be used to fix the edges of the floor plate member 500 to the floor.

[0105] (2) Detachable Weight

The above detachable weights may each be a metal composed of iron, brass, bronze, and/or copper or may also be composed of concrete. Any kinds of weights may be allowed as long as they can resist reaction force of a pedal of interest. For some pianos, two 5-kg weights or 20 four-hundred-gram metal sheets are sufficient for this purpose. The number of the weights are also not limited as long as the housing includes the weights. For this purpose, the housing may have a plurality of rooms for the weights. It may be preferable from the viewpoint of portability of the housing to provide separate weights rather than to provide one heavy weight.

[0106] (3) Sound-proof Chamber

When the actuator is actuated in response to the detector signal, the actuator-associated sound (noise) is generated. To minimize or cancel the noise, this housing has a sound-proof chamber for the actuator. The chamber may have a sound absorber on walls. The sound absorber may be acoustic foam. The acoustic foam is open celled foam used for acoustic treatment. It attenuates airborne sound waves by increasing air resistance, thus reducing the amplitude of the waves. The acoustic foam can be attached to walls, ceilings, doors, and other features of a room to control noise levels, vibrations, and echoes. Examples of materials for the sound absorber include polyurethane and glass wool.

[0107] Embodiment 5

Method for Operating Pedal

An embodiment of the present invention provides a method for operating a pedal, comprising the steps of: detecting a motion of a user to generate a detection signal; processing the detection signal to generate an actuator-controlling signal; and controlling a pedal in accordance with the actuator-controlling signal, wherein the motion involves a head movement; the head movement has an offset range within which the actuator is not actuated; and the actuator-controlling signal is changeable in accordance with the user's preference. The above described technical features through Embodiment 1 to Embodiment 4 are likewise applicable to this embodiment.

[0108] Embodiment 6

bFaaaP without Actuator

An embodiment of the present invention provides an auxiliary pedal system comprising: a detector configured to detect a motion of a user to generate a detection signal; and a controller configured to process the detection signal to generate and transmit an electric device-controlling command signal that replaces a pedal-mediated command signal, wherein the motion involves a head movement; the head movement has an offset range within which the electric device does not respond to the head movement; and the electric device-controlling command signal is changeable in accordance with the user's preference.

[0109] In this embodiment, the controller processes the detection signal to generate and transmit an electric device-controlling command signal that replaces a pedal-mediated command signal. This embodiment does not have an actuator configured to actually control a pedal. Instead, in this embodiment, the electric device-controlling command signal replaces a command signal generated by a pedal of the electric device and is input directly into an electric device controller for performing a function of interest. Here, the controller may include a detector-side control unit and an electric device-side control unit. The detector-side control unit alone or with the electric device-side control unit may process the detector signal in accordance with the features of the present invention. The controller according to this embodiment preferably includes this detector-side controller and may transmit the electric device-controlling command signal in a wired or wireless manner. This extends the present invention to a variety of electric devices with or even without any pedal. In this embodiment, the above-described features may be applicable.

[0110] Embodiment 7

Electric Piano

According to an embodiment of the present invention, the electric device may be an electric piano. Because each electric piano has an electric piano controller, the present invention may have a greater value for the electric piano than mechanical pianos. Just sending the electric device-controlling command signal to the electronic piano controller allows for an efficient way of enhancing piano performance by, in particular, the challenged with leg disabilities and kids.

[0111] Embodiment 8

Method for Operating Electric Device

An embodiment of the present invention provides a method for operating an electric device, comprising the steps of: detecting a motion of a user to generate a detection signal; and processing the detection signal to generate and transmit an electric device-controlling command signal that replaces a pedal-mediated command signal, wherein the motion involves a head movement; the head movement has an offset range within which the electric device does not respond to the head movement; and the electric device-controlling command signal is changeable in accordance with the user's preference. In this embodiment, the above-described features may be applicable so as to provide an efficient and universal way of controlling an electric device by almost everyone including the challenged with leg disabilities and kids.

[0112] Embodiment 9

Gaming Controller

An embodiment of the present invention provides a device controller comprising: a detector configured to detect a motion of a user to generate a detection signal; and a controller configured to process the detection signal to generate and transmit an electric device-controlling command signal, wherein the motion involves a head movement; the head movement has an offset range within which the electric device does not respond to the head movement; the electric device-controlling command signal is changeable in accordance with the user's preference; and the detector is attached to a side frame of eyeglasses.

[0113] This device controller may be a gaming controller. In a sense, playing the piano may be something like playing a game for some people. Use of a head motion may provide another way of inputting a user's action. Because this device controller is controlled by two elements involving an offset range and a multiplier changeable in accordance with the user's preference. This may provide another way of controlling a game play. Because almost everyone including the challenged with leg disabilities and kids can use this controller, a variety of games can be played by a wide range of gamers in a novel control way. Besides, this device controller may extend play control actions so as to provide a novel type of game.

Examples

[0114] Example 1

Overview of bFaaaP4

A bFaaaP4 is an auxiliary piano pedal system and was manufactured while the following points were taken into consideration:

1) A sustain pedal of a piano is depressed while an upper body movement, namely a head motion, is detected.

2) Operation of the system does not prevent a player from playing the piano and singing a song during performance.

3) The bFaaaP4 can be relatively easily constructed.

[0115] Example 1-1

Configuration

The bFaaaP4 was configured to include an angle sensor (1), a data processor (2), a transmitter (3), a receiver (4), an actuator controller (5), an actuator (6), and detector-side and actuator-side power sources (7) as shown in FIGS. 16 to 22. FIGS. 16 to 22 correspond to FIGS. 9 to 15, respectively, and the sizes of respective parts are represented in mm.

[0116] Angle Sensor

First, a 6-axis sensor (a combination of an accelerometer and a gyroscope) and a 9-axis sensor (a combination of an accelerometer, a gyroscope, and a geomagnetic sensor) were tested as the angle sensor. When a tilt angle is detected, the 9-axis sensor had less drifts and was commercially available and relatively inexpensive. Thus, an MPU9250 (manufactured by TDK Inc.) was used as the angle sensor. Because the MPU9250 was provided as a commercially available module, the module was enclosed in a case. The case was then detachably attached to a side frame of eyeglasses.

[0117] Data Processor

An ESP32 (manufactured by Espressif Systems, Inc.) was used as the data processor because it is relatively easy to develop a program. The data processor was equipped with an adjuster 1210 for an offset value (i.e., an upper limit of the above offset range) and an adjuster 1220 for a multiplier as shown in FIG. 3.

[0118] Transmitter and Receiver

The transmitter used Bluetooth BLE for communication with the receiver.

[0119] Actuator Controller

An ESP32 (manufactured by Espressif Systems, Inc.) was used as the actuator controller because it is relatively easy to develop a program. The actuator controller was equipped with an adjuster for a free pedal play offset value and an adjuster for an actuation range value as shown in FIG. 2.

[0120] Actuator

An electric cylinder (EAC4W-D05-AZAAD-1-G; manufactured by Oriental Motor Co., Ltd) was used as the actuator and DC24V AZ-Series α-step was used as a driver (EAC4-D05-AZAAD-1). Also, MEXE02 (Oriental Motor Co., Ltd) was used as set-up software.

[0121] Example 1-2

Detector-side Controller Program

The detector-side controller program was created in accordance with the detector-side processing flowcharts of FIGS. 4 to 5 by using Arduino IDE 1.8.5 with proper drivers installed therein. The pitch angle data was subjected to Madgwick filter processing to reduce noise and drifts and further subjected to offset processing and multiplier processing. Then, the processed value was converted to a control value ranging from 0 to 99. This control value was transmitted through BLE to the actuator controller.

[0122] Example 1-3

Actuator-side Controller Program

The actuator-side controller program was created in accordance with the actuator-side processing flowcharts of FIGS. 6 to 7 by using Arduino IDE 1.8.5 with proper drivers installed therein. The control value received at the actuator controller through the receiver was converted to an actuator command value (actuation width) set in view of a free pedal play offset value and an actuation range value so as to control and operate the actuator.

[0123] Example 1-4

Construction of Portable Housing with Actuator (Stepper Motor)

The bFaaaP4 housing was constructed in accordance with Embodiment 4. The size of each part or member of the housing was given in FIGS. 16 to 22. Each part and each member (each plate member was made of wood and colored black) were prepared and assembled by a routine procedure in the art (with each screw drilled into the wood plate member by using an electric screw driver). The actuator was the above linear stepper motor (manufactured by Oriental Motor Co., Ltd.). Also, 20 four-hundred-gram metal weights were arranged in two separate weight chambers (each chamber had the 10 weights). The actuator controller and the driver were put into the actuator controller chamber. The bottom side edges of the housing were attached to the floor by using a regular adhesive tape so as to further stabilize the housing.

[0124] Example 1-5

How to Operate bFaaaP4

The bFaaaP4 was operated in accordance with the operation protocol described in Embodiment (1-D-2).

[0125] Example 2

bFaaaP4W

In a bFaaaP4W, a cable was used for the communication between the transmitter and the receiver. Thus, no BLE communication was used because some reports indicated malfunction of BLE in ESP32. The control programs were substantially the same as of the bFaaaP4. Note that the actuator response to a user’s head motion was improved partly because of an increased communication speed between the transmitter and the receiver.

[0126] Example 3

bFaaaP3

A bFaaaP3 had substantially the same configuration as of the bFaaaP4. In order to avoid redundancy, differences from the bFaaaP4 are described.

There were three major differences therebetween. The first difference involved the detector-side controller. The detector-side controller of the bFaaaP3 had substantially the same configuration as of the bFaaaP4, but the adjuster for a head angle tilt angle offset value and the adjuster for a multiplier did not each have a visible scale of an adjuster button. Thus, the offset value and the multiplier were able to be adjusted by each user, but the actual value could not be determined with precision.

The second difference involved the sound-proof chamber. Although the bFaaaP4 had a sound-proof chamber that housed the actuator, the actuator of the bFaaaP3 was attached to a side wall member by using bolts and nuts and was exposed to the outside without a cover. So, the actuator-derived noise was not attenuated.

The third difference involved the weights. Instead of using 20 four-hundred-gram metal plates, 2 five-kg cement blocks were placed in the weight chambers.

The operation protocol was identical to that of the bFaaaP4.

[0127] Example 4

bFaaaP2

A bFaaaP2 had substantially the same configuration as of the bFaaaP3. In order to avoid redundancy, differences from the bFaaaP3 are described.

There was one major difference therebetween. This difference involved the detector-side controller. As the angle sensor and the data processor was used a M5Stack_Gray (purchased from SWITCH SCIENCE) including an MPU9250. The program was substantially the same as that of the bFaaaP3 or the bFaaaP4. This M5Stack was put into a pocket attached to a cap so as to place the M5Stack on the head of each user wearing the cap. Although attached to the head, it was difficult to place the angle detector horizontally so as to detect a head tilt angle with precision. A head tilt angle offset value and a multiplier were easily input by using buttons of the M5Stack.

[0128] Example 5

bFaaaP1

A bFaaaP1 had substantially the same configuration as of the bFaaaP2. In order to avoid redundancy, differences from the bFaaaP2 are described.

There were three major differences therebetween. The first difference involved the actuator. The actuator of the bFaaaP1 was a stepper motor VESTA UOK5141 (manufactured by Oriental Motor Co., Ltd.), which is not a linear cylinder. This stepper motor was operated from the right side of a piano.

The second difference involved the actuator controller. The actuator controller of the bFaaaP1 was an ESP32 on an Arduino board (http://akizukidenshi.com/catalog/g/gM-11819/). The adjuster for a free pedal play offset value and the adjuster for an actuation range value were substantially the same as of the bFaaaP2.

The third difference involved a program for the detector-side controller. As the detector-side controller was used the above M5Stack. However, the program was to detect a Z-axis rotation rate of the head of each user to generate a detector signal. The detector signal was converted to a binary control value so as to either depress or release a pedal. A head motion offset range was set in the program. But a multiplier was not set by each user.

[0129] Example 6

Auxiliary Pedal Effect Evaluation Study (APEE study)

The present auxiliary pedal systems were evaluated on Class I to III subjects. Here, 7 Class I subjects, 5 Class II subjects, and 3 Class III subjects were enrolled in this auxiliary pedal effect evaluation study (APEE study). The Class II subjects were students of Kyoko Yamaguchi piano class Fleur (https://www.fleur-pianok.com/).

[0130] Test Procedure

First, all the subjects were instructed to read a piano pedal test protocol and practice the piano notes in the sheet music of FIG. 23 (i.e., notes consist of 4 repeat units of Do, Do, Do, Re, Re, Re, and Mi, Mi, Mi).

[0131] Test protocol was provided as follows:

1. Read a test sheet music and image a pedal operation

2. Practice notes of the test sheet music without using a bFaaaP

3. Practice the notes while using the bFaaaP.

4. Adjust an offset value and a multiplier and select them according to your preference (for class II subjects (kids), the offset value was set to 5 degrees and the multiplier was set to 20 or 30).

5. Start a recording and play the notes.

5-1. Say your name and the date of this recording and the offset value and the multiplier you selected.

5-2. Play the notes without using a pedal

5-3. Play the notes with pedal pattern 1 by using the bFaaaP.

5-4. Play the notes with pedal pattern 2 by using the bFaaaP.

5-5. End the recording.

6. Fill in a questionnaire.

[0132] The questionnaire is shown in FIG. 24.

[0133] An upright piano K132 (Production No. 599580; manufactured in Hamburg by Steinway&Sons), a grand piano GC1 (Production No. 6191885; manufactured in 2007 by YAMAHA), and an upright piano UX (Production No. 2424245; manufactured in 1977 by YAMAHA) were used for this APEE study. A sustain pedal was depressed in accordance with pedal pattern 1 or 2 by using a bFaaaP (bFaaaP2, bFaaaP3, bFaaaP4, or bFaaaP4W). Note that each bFaaaP was arranged on the right front side of the sustain pedal. Next, each subject chose a head tilt angle offset value and a multiplier by adjusting a lever and a dial of the detector-side controller and the chosen offset value and multiplier were recorded. Then, the notes were played with or without pedal actions controlled by the bFaaaP. The piano performance was recorded as a wav file in a smartphone (iPhone). As a control, Class I subjects were requested to play the notes by their own foot. After the test, each subject filled in a test sheet and gave their comments as well as a written non-disclosure consent.

[0134] Analysis

Each wav file was incorporated in Sonic Visualiser Release 3.0.3 and an area of interest was exported as a png file. Each png file was then opened in ImageJ 1.51j8. Its image type and threshold were set to 8 bit and 200, respectively. After each of note portions with or without pedal actions (i.e., no pedal, pattern 1 or pattern 2) was selected, the area of each portion was calculated as a tone-vibrating area (TVA) by the "Analyze Particles" command. Data of Test No. 17 is shown in FIG. 25A. When a certain note portion contained more than 4 repeat units, the first 4 repeat units were used for analysis. Also, when a certain note portion contained only 3 repeat units, only 3 repeat units were used for the other note portions to calculate TVAs. Subsequently, the area of each portion (i.e., TVA0, TVA1, and TVA2) was compared while the area of the portion without pedal actions (i.e., TVA0) was set to 1 and the other areas with pedal actions (pattern 1 or pattern 2) (i.e., TVA1 or TVA2) were each provided as a score (ratio relative to TVA0) as shown in FIG. 25B.

[0135] Results and Statistics

Table 1 shows the results of Class I subjects in the APEE study. Table 2 shows the results of Class II and III subjects.

[0136]

[Table 1]

Auxiliary Pedal Effect Evaluation Study (APEE study)

Subject Class	Subject Number	Test Date	Age	Piano Experience	Piano	Device (with or w/o bFaaaaP)	Offset Va ue	Multiplier	Pattern 0	Pattern 1	Pattern 2	Test No.
									Tone-vibrating Area (TVA)			
									Relative to Pattern 0			
Class I (adults)	No.1	2018/8/26	50	1 to 2 years (when he was a child)	K132	Regular Pedal	N.A.	N.A.	1.00	1.26	1.81	1
						bFaaaP2 (M5Stack)	5	15	1.00	1.05	1.26	2
						bFaaP3	N.D.	N.D.	1.00	1.15	1.27	3
	No. 2	2018/9/19	66	0 years	K132	bFaaaP4	19	29	1.00	1.86	3.30	4
	No. 3	2018/9/19	53	0 years	K132	bFaaaP4	10	30	1.00	1.41	1.78	5
		2018/10/7			UX	Regular Pedal	N.A.	N.A.	1.00	1.29	1.31	6
						bFaaaP4	15	25	1.00	1.30	1.58	7
							10	40	1.00	1.25	1.45	8
		2018/10/31			K132	Regular Pedal	N.A.	N.A.	1.00	1.29	1.51	9
		2018/10/31			GP1	bFaaaP4W	5	20	1.00	1.67	1.97	10
							5	30	1.00	1.70	1.78	11
						Regular Pedal	N.A.	N.A.	1.00	1.41	1.41	12
	No. 4	2018/9/23	48	Many years	K132	bFaaaP4	5	20	1.00	1.77	1.97	13
							10	20	1.00	1.63	1.70	14
							5	40	1.00	1.80	1.94	15
						Regular Pedal	N.A.	N.A.	1.00	1.90	2.46	16
	No. 5	2018/9/27	35	Many years	K132	bFaaaP4	10	30	1.00	1.64	1.83	17
							15	30	1.00	1.73	1.83	18
						Regular Pedal	N.A.	N.A.	1.00	1.92	2.06	19
		2018/10/31			GP1	bFaaaP4W	5	20	1.00	1.94	1.97	20
							5	30	1.00	1.82	1.93	21
						Regular Pedal	N.A.	N.A.	1.00	1.88	2.20	22
	No. 6	2018/10/7	In her teens	3 years	UX	Regular Pedal	N.A.	N.A.	1.00	1.18	1.45	23
		2018/10/8				bFaaaP4	5	10	1.00	1.43	1.48	24
							10	30	1.00	1.19	1.28	25
	No. 7	2018/10/7	In her fourties	6 years	UX	Regular Pedal	N.A.	N.A.	1.00	1.08	1.06	26
						bFaaaP4	10	30	1.00	1.27	1.27	27
							5	10	1.00	1.12	1.17	28

[0137]

[Table 2]

Auxiliary Pedal Effect Evaluation Study (APEE study)

Subject Class	Subject Number	Test Date	Age	Piano Experience	Piano	Device (with or w/o bFaaaaP)	Offset Value	Multiplier	Pattern 0	Pattern 1	Pattern 2	Test No.
									Tone-vibrating Area (TVA)			
									Relative to Pattern 0			
Class II (kids)	No. 8	2018/10/31	11	2 years and 7 months	GP1	bFaaaP4W	5	20	1.00	1.83	2.19	29
							5	30	1.00	2.05	2.23	30
	No. 9	2018/10/31	8	1 year and 6 months	GP1	bFaaaP4W	5	20	1.00	1.87	1.78	31
							5	30	1.00	1.19	1.01	32
	No. 10	2018/10/31	10	5 years	GP1	bFaaaP4W	5	20	1.00	2.17	2.42	33
							5	30	1.00	1.66	1.75	34
	No. 11	2018/10/31	8	2 years	GP1	bFaaaP4W	5	20	1.00	2.18	2.57	35
							5	30	1.00	1.56	1.56	36
	No. 12	2018/10/31	7	2 years	GP1	bFaaaP4W	5	20	1.00	2.06	1.89	37
							5	30	1.00	1.23	1.68	38
Class III (the challenged)	No. 13 (with leg disabilities)	2018/8/26	16	0 years	K132	BFaaaP2 (M5Stack)	10	15	1.00	1.09	0.90	39
						bFaaaP3	N.D	N.D	1.00	1.32	1.30	40
	No. 14 (with partial leg disabilities and trachectomy)	2018/9/23	19	Several years	K132	bFaaaP4	5	40	1.00	1.18	1.36	41
							3	50 (Max)	1.00	0.98	1.67	42
	No. 15 (with leg disabilites)	2018/8/5	50	5 years	K132	bFaaaP2 (M5Stack)	5	8	1.00	1.72	2.12	43
		2018/9/19				bFaaaP4	10	30	1.00	1.84	2.22	44
		2018/9/23					10	40	1.00	1.70	1.79	45
		2018/9/27					5	40	1.00	1.56	2.17	46

[0138] When the bFaaaP3 was used in the tests, the offset value and the multiplier were not determined (N.D.) in Tables 1 and 2. In the bFaaaP3, they were adjustable, but precise values were not indicated because there was no scale displayed. The term "N.A." in Table 1 means "not applicable" because relevant users used their own foot to depress a pedal. In Test Nos. 2, 39, and 43, the bFaaaP2 (M5Stack) was used. Although Subject No. 15 gave a good score (i.e., a relative ratio), the other two scores were relatively low. This may be because the bFaaaP2 used an M5Stack and the M5Stack was put into a pocket attached to a cap, so that it was more difficult to keep an offset angle than when the detector was attached to a side frame of eyeglasses. In addition, how much one practiced the bFaaaP might affect the scores. Thus, in the following statistics, the results of using the bFaaaP2 were excluded from the statistics. The results of using a bFaaaP3, a bFaaaP4, and a bFaaaP4W were used as bFaaaP scores in the following statistics.

[0139] Differences in Scores between Pedal Patterns Played with or without bFaaaP

The results of pedal pattern 0 (without using a pedal) were compared with the results of pedal pattern 1 or 2 (as obtained by using each bFaaaP). First, a two-tailed paired t-test was conducted on scores (relative ratio) between pedal pattern 0 and pedal pattern 1. Next, a two-tailed paired t-test was conducted on scores between pedal pattern 0 and pedal pattern 2. Then, a two-tailed paired t-test was conducted on scores between pedal pattern 1 and pedal pattern 2. Table 3 shows the results.

[0140] [Table 3]

Scores	Average	Standard Deviation
Pedal Pattern 0	1.00	0
Pedal Pattern 1 Using bFaaaP	1.59	0.32
Pedal Pattern 2 Using bFaaaP	1.80	0.44
Score Comparison	p-Value	
Pattern 0 vs. Pattern 1	4.64 E-12 (< 0.01)	
Pattern 0 vs. pattern 2	7.25 E-12 (< 0.01)	
Pattern 1 vs. pattern 2	0.00023 (< 0.01)	

[0141] Table 3 shows that the score of pedal pattern 1 was 1.50 ± 0.32 (average ± S.D.) and the score of pedal pattern 2 was 1.80 ± 0.44 while the score of pedal pattern 0 was set to 1.00. The scores between pedal pattern 0 and pedal pattern 1 and between pedal pattern 0 and pedal pattern 2 were significantly different (each p-Value was < 0.01), demonstrating that each bFaaaP was effective in suspending tones when used by the study subjects. In addition, the scores between pedal pattern 1 and pedal pattern 2 were also significantly different (p = 0.00023 (< 0.01)), suggesting that different pedal patterns produced different effects.

[0142] bFaaaP vs. Leg

To examine whether or not the scores obtained by using a bFaaaP differed from those obtained by using their own leg, these scores were first subject to an F-test, in which whether two groups have a statistically equal variance is tested, and then subject to a t-test, in which whether the two groups are significantly different is tested. First, scores (relative ratios) of pedal pattern 1 or 2 using each bFaaaP and scores of pedal pattern 1 or 2 using their own leg were subject to the F-test. The results show that the null hypothesis is not rejected (p > 0.05), so that statistically equal valiance between both could be assumed. Then, a two-tailed t-test assuming equal variance was conducted. Table 4 shows the results.

[0143] [Table 4]

Scores	Average (the number of samples)	Standard Deviation
Pedal Pattern 1 Using bFaaaP	1.59 (34)	0.32
Pedal Pattern 1 Using Leg	1.47 (9)	0.32
Pedal Pattern 2 Using bFaaaP	1.80 (34)	0.44
Pedal Pattern 1 Using Leg	1.70 (9)	0.44
Comparison	F-Test p-Value	
Pedal Pattern 1 bFaaaP vs. Leg	0.84 (> 0.05)	
Pedal Pattern 2 bFaaaP vs. Leg	0.83 (> 0.05)	
Comparison	t-Test p-Value	
Pedal Pattern 1 bFaaaP vs. Leg	0.33 (> 0.05)	
Pedal Pattern 2 bFaaaP vs. Leg	0.43 (> 0.05)	

[0144] Table 4 shows that there was no statistically significant difference (p > 0.05) in scores between each bFaaaP and their own leg, suggesting that each bFaaaP can exert substantially the same effects obtained by using their own leg.

[0145] Did Piano Experience Matter?

In this study, different subjects had different piano experiences. To check whether or not piano experience of each subject affected scores, scores of subjects having piano experience of 5 years or more (Group I) were compared with those having piano experience of less than 5 years (Group II). First, scores of Group I and scores of Group II were subject to the F-test. The results show that the null hypothesis is not rejected ($p > 0.05$), so that statistically equal valiance between both could be assumed. Then, a two-tailed t-test assuming equal variance was conducted. Table 5 shows the results.

[0146] [Table 5]

Scores	Average (the number of samples)	Standard Deviation
Pedal Pattern 1 Group I	1.69 (14)	0.25
Pedal Pattern 1 Group II	1.52 (20)	0.35
Pedal Pattern 2 Group I	1.85 (14)	0.32
Pedal Pattern 1 Group II	1.76 (20)	0.51
Comparison	F-Test p-Value	
Pedal Pattern 1 Group I vs. Group II	0.24 (> 0.05)	
Pedal Pattern 2 Group I vs. Group II	0.10 (> 0.05)	
Comparison	t-Test p-Value	
Pedal Pattern 1 Group I vs. Group II	0.14 (> 0.05)	
Pedal Pattern 2 Group I vs. Group II	0.54 (>0.05)	

[0147] Table 5 shows that there was no statistically significant difference ($p > 0.05$) in scores between Group I and Group II, suggesting that the piano experience did not matter statistically at least in this APEE study.

[0148] Inter-class Differences

To examine whether or not there was a difference among the subject classes, one-way ANOVA (analysis of variance) was conducted. Table 6 shows the results.

[0149] [Table 6]

Pedal Pattern 1			
Class Name	The Number of Data	Average	Variance
Class I	18	1.54	0.074
Class II	10	1.78	0.13
Class III	6	1.43	0.10
Analysis	Degree of Freedom	P-value	F-value
Inter-group	2	0.068 (> 0.05)	3.30
Pedal Pattern 2			
Class Name	The Number of Data	Average	Variance
Class I	18	1.75	0.23
Class II	10	1.91	0.21
Class III	6	1.75	0.15
Analysis	Degree of Freedom	P-value	F-value
Inter-group	2	0.66 (> 0.05)	3.30

[0150] Table 6 shows that there was no statistically significant difference among Classes I, II, and III with respect to both the pedal patterns 1 and 2 (i.e., the null hypothesis was not rejected because both the p values were more than 0.05), suggesting that the pedal-sustaining effects exerted by using each bFaaaP had no statistically significant differences among the Classes.

[0151] Inter-piano Differences

To examine whether or not there was a difference among the pianos used, one-way ANOVA (analysis of variance) was conducted. Table 7 shows the results.

[0152]

[Table 7]

Pedal Pattern 1			
Piano Name	The Number of Data	Average	Variance
K132	14	1.54	0.081
GP1	14	1.78	0.095
UX	6	1.26	0.011
Analysis	Degree of Freedom	P-value	F-value
Inter-group	2	0.0012 (< 0.01)	3.30
Pedal Pattern 2			
Piano Name	The Number of Data	Average	Variance
K132	14	1.87	0.25
GP1	14	1.90	0.15
UX	6	1.37	0.024
Analysis	Degree of Freedom	P-value	F-value
Inter-group	2	0.032 (< 0.05)	3.30

[0153] Table 7 shows that there was a statistically significant difference among the pianos with respect to both the pedal patterns 1 and 2 (i.e., the null hypothesis was rejected because one of the p values was less than 0.01 and the other p value was less than 0.05), suggesting that the pedal-sustaining effects exerted by using each bFaaaP had a statistically significant difference among the pianos.

[0154] Questionnaire

All the subjects were requested to fill in the form of FIG. 24. Table 8 shows the results. The number of votes was indicated below each evaluation criterion about the impressions.

[0155] [Table 8]

Impression		Good	Intermediate	Poor	Total
1	Did you enjoy it?	9	5	1	15
2	Easy to use?	3	10	2	15
3	Good appearance?	4.5	7.5	3	15
4	Does the name of bFaaaP sound good?	8	6	1	15

[0156] The relatively large number of study subjects enjoyed the bFaaaP experience and found the naming of the device good. Also, more than half of the subjects found that the bFaaaP was easy to use and had a good appearance.

[0157] Comments of Study Subjects

Some of comments of the study subjects are provided as follows.

(1) The eyeglasses-attached sensor is compact and good.

(2) There is some actuator noise, but the noise may let a user know the pedal ON/OFF timing.

(3) The pedal-releasing operation is more difficult than the pedal-depressing operation.

(4) The actuator follows the head movement more naturally than I thought.

(5) Use of horizontal movement of the head may be an option.

(6) I think that the more you practice, the more easily you can use the device.

[0158] Discussion

The bFaaaPs were demonstrated to be effective in sustaining a tone as proved in Table 3. Also, different pedal patterns played using each bFaaaP caused statistically different scores accordingly (i.e., pedal pattern 1 vs. pedal pattern 2 ($p = 0.00023$)), suggesting precise pedal control executed by each bFaaaP. Further, the scores obtained by using each bFaaaP were not significantly different from the scores obtained by using their own leg (i.e., the respective p-values are 0.33 and 0.43 (> 0.05)), suggesting equivalent pedal control between each bFaaaP and their own leg (see Table 4). However, each TVA varies depending on subjects. This may be because the subjects had preferred piano key touches. In addition, how much one practiced the bFaaaP might affect the scores.

[0159] The user-selected offset angle value ranged from 3 to 19 degrees while 5 or 10 degrees were preferred by many users (see Tables 1 and 2). Also, the user-selected multiplier ranged from 8 to 50 while a range of 10 to 50 was selected by the users when the bFaaaP4 was used. Also, a range of 20 to 40 was preferred by many users. That is, in many cases, the users selected 5 to 10 degrees as an offset angle and additional 2.5 to 5 or 10 degrees from the offset angle were used to fully depress a pedal.

[0160] In the case of Subject No. 14, she had partial leg disabilities and tracheotomy and selected 5 or 3 as the offset value and 40 or 50 (maximum) as the multiplier, respectively. This is because her head movement was restricted due to the tracheotomy. The smaller the offset value and the larger the multiplier, the better for her. By setting both the values as described above, she was able to better use the bFaaaP to play the above notes as demonstrated in the case of the pedal pattern 2 of Test No. 42.

[0161] Here, in order to provide precise control of the pedal, a smaller multiplier is better because the pedal actuation width can be controlled by a larger range of the head tilt angle. However, a quicker pedal action may be realized by using a greater multiplier. Also, the offset value varies depending on subjects because the different subjects have different offset values in accordance with their preferences. That is, some want to vigorously move their head; and others want to slightly move their head. The present bFaaaP enables the pedal operation in either way by selecting the offset value and the multiplier. The effects of the bFaaaP are not regarded as an additive effect exerted by the offset value and the multiplier, but a synergistic effect exerted by a combination of the offset value and the multiplier as demonstrated above.

[0162] In this APEE study, there were no statistically significant differences (i.e., each p-value was more than 0.05) in scores between Group I (subjects having piano experience of 5 years or more) and Group II (subjects having piano experience of less than 5 years) (see Table 5) as well as there were no statistically significant differences (i.e., each p-value was more than 0.05) in scores among the subject classes (i.e., Class I: adults (those who can depress a pedal by their own leg); Class II: kids; and Class III; the challenged) (see Table 6). This indicates that bFaaaPs can be used irrespective of the piano experience and among various persons including kids and the challenged. However, the p-value obtained by the one-way ANOVA for pedal pattern 1 among the classes was 0.068, suggesting a potential difference among the classes.

[0163] Meanwhile, Table 7 shows that there were statistically significant differences in scores among the pianos used. In this one-way ANOVA, the p-value for pedal pattern 1 was 0.0012 (< 0.01) and the p-value for pedal pattern 2 was 0.032 (< 0.05). When the respective score averages are compared, the K132 and the GP1 exhibited substantially the same values. So, the differences may be due to the UX. It is unclear what caused the differences. One factor may be the structure and the damper mechanism of each piano. Another factor may be tuning. While tuning of the piano was regularly conducted on the K132 and the GP1, tuning of the UX was not performed for more than 8 years.

[0164] Overall, bFaaaPs can be used for a variety of users as an alternative for their own leg and make it possible to precisely control a pedal in accordance with the user's preference. This idea and concept of the detector and the controller (processor) are applicable to any electric devices with or even without a pedal, in particular, electric pianos and gaming controllers.

[0165] <u>Example 7</u>

<u>Free Pedal Play Measurement</u>

The free pedal play of each of the above pianos used herein was measured with a digital caliper (manufactured by Shinwa Rules Co., Ltd.). The point where a note starts sustaining by depressing a pedal was determined manually. The distance from the initial pedal position to the above point was measured and set as a free pedal play represented in mm. Table 9 shows the results.

[0166]

[Table 9]

Piano	Free pedal play	Measurement date (YYYY/MM/DD)
K132 (an upright piano, Steinway&Sons)	19 mm	2018/09/19
GC1 (a grand piano, YAMAHA)	14 mm	2018/10/31
UX (an upright piano, YAMAHA)	20 mm	2018/10/10

[0167] The results show that the three pianos used herein had different free pedal play values. Thus, it is necessary to adjust and initialize each bFaaaP in accordance with each piano.

[0168] Example 8

Reaction Force Measurement

Reaction force of a pedal of each of the above pianos used herein was measured manually. For the measurement, a mechanical force gage NK-100 (push pull gage; manufactured by Hanchen JP) was used in accordance with the manufacturer's instruction. The pedal force when the pedal was depressed to the above free pedal play point and the pedal force when the pedal was fully depressed were measured. Table 10 shows the results.

[0169] [Table 10]

Piano	Reaction force at free pedal play point	Reaction force when fully depressed	Measurement date (YYYY/MM/DD)
K132 (an upright piano, Steinway&Sons)	4.0 kg	5.5 kg	2018/09/19
GC1 (a grand piano, YAMAHA)	3.0 kg	8.0 kg	2018/10/31
UX (an upright piano, YAMAHA)	2.2 kg	4.1 kg	2018/10/10

[0170] Table 10 shows that the reaction forces differed from one piano to another. This difference may be due to the structure of a damper mechanism. However, the maximum was 8.0 kg. Accordingly. the sufficient weights, which are installed in the housing, may be 8 to 10 kg or more.

[0171] Example 9

Measurement of Head Tilt Angles during Piano Performance

The second movement of Piano Sonata No. 8 in C minor, Op. 13 (Sonata Pathetique) composed by Ludwig van Beethoven was played by Class III Subject No. 15 on September 19, 2018. The piano performance was analyzed by the above bFaaaP controller program as follows: (1) head tilt angles during the piano performance were detected by the bFaaaP4; (2) the tilt angles were sequentially output through a USB port to a PC; (3) the resulting data was statistically analyzed. Table 11 and FIG. 26 show the results.

[0172] [Table 11]

Angle Range (to the following angle)	Frequency (events)	Percentage (%)
-20	0	0
-17.5	0	0
-15	7	0
-12.5	20	0
-10	44	0
-7.5	348	0
-5	1442	1
-2.5	7275	5
0	28001	19
2.5	53083	36
5	37356	25
7.5	12994	9
10	5069	3
12.5	907	1
15	394	0
17.5	150	0
20	109	0
	0	
Total	147199	100

[0173] Results

The maximum tilt angle was 19.13 degrees and the minimum tilt angle was -15.02 degrees. The histogram analysis of Table 11 and FIG. 26 demonstrated that the tilt angle range equal to or less than 2.5 degrees accounted for 61% of all the detection events recorded; the tilt angle range equal to or less than 5.0 accounted for 86%; the tilt angle range equal to or less than 7.5 accounted for 95%; and the tilt angle range equal to or less than 10.0 accounted for 99%. Thus, from the viewpoint of voluntary head movements, the head tilt angle offset range is set to preferably 2.5 degrees (3 degrees) or less, more preferably 5.0 degrees or less, still more preferably 7.5 degrees or less, and still more preferably 10.0 degrees or less.

[0174] All the Drawings were designed and created by Ootaki Architect & Craftsmen (https://anity.ootaki.info/) (also referred to as AnITy Design; https://scrapbox.io/ootaki/). All the programs cited herein will be able to be seen in the bFaaaP website, which will be opened after the filing of this patent application.

[0175] The above-described Embodiments and Examples are just examples in the present invention. Accordingly, they should not be construed such that the technical scope of the present invention is limited. This is because the present invention can be put into practice, without departing from the spirit and the main features thereof, even in various embodiments.

Industrial Applicability

[0176] The present invention is applicable to any devices and musical instruments (e.g., a piano) with a pedal. Also, the present invention is applicable to any electric devices (e.g., an electric piano, a gaming controller) and can be used universally by almost everybody including the challenged with leg disabilities and kids.

Reference Signs List

[0177] 100 Housing
200 Sound-proof chamber
210 Sound Absorber
250 Opening
300 First weight chamber
310 Bolt
350 Second weight chamber
360 Bolt
500 Floor plate member
510 Sound-proof chamber bottom plate member
520 First horizontal plate member
530 Second horizontal plate member
540 Ceiling plate member
550 First side plate member
560 Second side plate member
570 Third side plate member
580 Rear plate member
590 Front plate member
592b1 to b2 Inner rubber plug

592a1 to a2 Outer rubber plug
594b1 to b2 Inner rubber plug
594a1 to a2 Outer rubber plug
595 Plug bolts
600 Actuator (Motor)
610 Body portion
615 Actuator attachment plate
620 Leg portion
630 Spacing plate member
700a1-n Weights
750b1-n Weights
800 Actuator controller chamber
810 Initializer (Adjusters for a free pedal play offset value and an actuation range value)
820 Actuator controller
830 Driver unit (Driver and power supply)
900 Vibration-proof sheet
1000 Detector (Sensor)
1010 Case
1020 Slit or groove
1050 Sensor module (MPU9250)
1100 Side frame of eyeglasses
1200 Detector-side controller
1210 Adjuster for an offset range value
1220 Adjuster for a multiplier
1230 On/off switch

Claims

[Claim 1] An auxiliary pedal system comprising:
a detector configured to detect a motion of a user to generate a detection signal;
a processor configured to process the detection signal to generate an actuator-controlling signal; and
an actuator configured to control a pedal in accordance with the actuator-controlling signal,
wherein the motion involves a head movement;
the head movement has an offset range within which the actuator is not actuated; and
the actuator-controlling signal is changeable in accordance with the user's preference.

[Claim 2] The auxiliary pedal system according to claim 1, wherein the detector is attached to eyeglasses.

[Claim 3] The auxiliary pedal system according to claim 2, wherein the detector is attached to a side frame of the eyeglasses and a case for the detector has a groove or slit fit for the side frame.

[Claim 4] The auxiliary pedal system according to claim 1, wherein the head movement involves a head tilt angle;
the offset range of the head tilt angle is defined by the user and an upper limit of the offset range is 10 degrees or less.

[Claim 5] The auxiliary pedal system according to claim 1, wherein the head movement involves a head tilt angle;
the offset range of the head tilt angle is defined by the user and an upper limit of the offset range is 7.5 degrees or less.

[Claim 6] The auxiliary pedal system according to claim 1, wherein the head movement involves a head tilt angle;
the offset range of the head tilt angle is defined by the user and an upper limit of the offset range is 5 degrees or less.

[Claim 7] The auxiliary pedal system according to claim 1, wherein the head movement involves a head tilt angle;
the offset range of the head tilt angle is defined by the user and an upper limit of the offset range is 3 degrees or less.

[Claim 8] The auxiliary pedal system according to claim 2, wherein the head movement involves a head tilt angle;
the offset range of the head tilt angle is defined by the user and an

upper limit of the offset range is 10 degrees or less.

[Claim 9] The auxiliary pedal system according to claim 2, wherein the head movement involves a head tilt angle;
the offset range of the head tilt angle is defined by the user and an upper limit of the offset range is 7.5 degrees or less.

[Claim 10] The auxiliary pedal system according to claim 2, wherein the head movement involves a head tilt angle;
the offset range of the head tilt angle is defined by the user and an upper limit of the offset range is 5 degrees or less.

[Claim 11] The auxiliary pedal system according to claim 2, wherein the head movement involves a head tilt angle;
the offset range of the head tilt angle is defined by the user and an upper limit of the offset range is 3 degrees or less.

[Claim 12] The auxiliary pedal system according to claim 4, wherein the actuator-controlling signal is modified by multiplying, by a multiplier, a value obtained by subtracting the upper limit of the offset range from the head tilt angle; and
the multiplier is selected by the user and from 10 to 50, so that 2 to 10 degrees of the head tilt angle from the upper limit of the offset range are used to fully depress the pedal by the actuator.

[Claim 13] The auxiliary pedal system according to claim 7, wherein the actuator-controlling signal is modified by multiplying, by a multiplier, a value obtained by subtracting the upper limit of the offset range from the head tilt angle; and
the multiplier is selected by the user and from 10 to 50, so that 2 to 10 degrees of the head tilt angle from the upper limit of the offset range are used to fully depress the pedal by the actuator.

[Claim 14] The auxiliary pedal system according to claim 8, wherein the actuator-controlling signal is modified by multiplying, by a multiplier, a value obtained by subtracting the upper limit of the offset range from the head tilt angle; and
the multiplier is selected by the user and from 10 to 50, so that 2 to 10 degrees of the head tilt angle from the upper limit of the offset range are used to fully depress the pedal by the actuator.

[Claim 15] The auxiliary pedal system according to claim 11, wherein the actuator-controlling signal is modified by multiplying, by a multiplier, a value obtained by subtracting the upper limit of the offset range from the head tilt angle; and the multiplier is selected by the user and from 10 to 50, so that 2 to 10 degrees of the head tilt angle from the upper limit of the offset range are used to fully depress the pedal by the actuator.

[Claim 16] The auxiliary pedal system according to claim 12, wherein the pedal is a piano pedal.

[Claim 17] The auxiliary pedal system according to claim 13, wherein the pedal is a piano pedal.

[Claim 18] The auxiliary pedal system according to claim 14, wherein the pedal is a piano pedal.

[Claim 19] The auxiliary pedal system according to claim 15, wherein the pedal is a piano pedal.

[Claim 20] The auxiliary pedal system according to claim 16, where a free pedal play is offset such that the pedal responds to the actuator-controlling signal and moves without a delay.

[Claim 21] The auxiliary pedal system according to claim 17, where a free pedal play is offset such that the pedal responds to the actuator-controlling signal and moves without a delay.

[Claim 22] The auxiliary pedal system according to claim 18, where a free pedal play is offset such that the pedal responds to the actuator-controlling signal and moves without a delay.

[Claim 23] The auxiliary pedal system according to claim 19, where a free pedal play is offset such that the pedal responds to the actuator-controlling signal and moves without a delay.

[Claim 24] The auxiliary pedal system according to claim 20, further comprising a portable housing having the actuator and a detachable weight, wherein the detachable weight resists reaction force of the actuator so as not to move the housing; and
the housing has a sound-proof chamber including the actuator.

[Claim 25] The auxiliary pedal system according to claim 21, further comprising a portable housing having the actuator and a detachable weight, wherein the detachable weight resists reaction force of the actuator so as not to move the housing; and
the housing has a sound-proof chamber including the actuator.

[Claim 26] The auxiliary pedal system according to claim 22, further comprising a portable housing having the actuator and a detachable weight, wherein the detachable weight resists reaction force of the actuator so as not to move the housing; and
the housing has a sound-proof chamber including the actuator.

[Claim 27] The auxiliary pedal system according to claim 23, further comprising a portable housing having the actuator and a detachable weight, wherein the detachable weight resists reaction force of the actuator so as not to move the housing; and
the housing has a sound-proof chamber including the actuator.

[Claim 28] A method for operating a pedal, comprising the steps of:
detecting a motion of a user to generate a detection signal;
processing the detection signal to generate an actuator-controlling signal; and
controlling a pedal in accordance with the actuator-controlling signal,
wherein the motion involves a head movement;
the head movement has an offset range within which the actuator is not actuated; and
the actuator-controlling signal is changeable in accordance with the user's preference.

[Claim 29] An auxiliary pedal system comprising:
a detector configured to detect a motion of a user to generate a detection signal; and
a controller configured to process the detection signal to generate and transmit an electric device-controlling command signal that replaces a pedal-mediated command signal,
wherein the motion involves a head movement;
the head movement has an offset range within which the electric device does not respond to the head movement; and
the electric device-controlling command signal is changeable in accordance with the user's preference.

[Claim 30] The auxiliary pedal system according to claim 29, wherein the detector is attached to eyeglasses and the head movement involves a head tilt angle;
the offset range of the head tilt angle is defined by the user and an upper limit of the offset range is 10 degrees or less;
the electric device-controlling command signal is modified by multiplying, by a multiplier, a value obtained by subtracting the upper limit of the offset range from the head tilt angle; and
the multiplier is selected by the user and from 10 to 50, so that 2 to 10 degrees of the head tilt angle from the upper limit of the offset range are used to fully operate the electric device.

[Claim 31] The auxiliary pedal system according to claim 29, wherein the detector

is attached to eyeglasses and the head movement involves a head tilt angle;
the offset range of the head tilt angle is defined by the user and an upper limit of the offset range is 3 degrees or less;
the electric device-controlling command signal is modified by multiplying, by a multiplier, a value obtained by subtracting the upper limit of the offset range from the head tilt angle; and
the multiplier is selected by the user and from 10 to 50, so that 2 to 10 degrees of the head tilt angle from the upper limit of the offset range are used to fully operate the electric device.

[Claim 32] The auxiliary pedal system according to claim 30, wherein the electric device is an electric piano.

[Claim 33] The auxiliary pedal system according to claim 31, wherein the electric device is an electric piano.

[Claim 34] A method for operating an electric device, comprising the steps of:
detecting a motion of a user to generate a detection signal; and
processing the detection signal to generate and transmit an electric device-controlling command signal that replaces a pedal-mediated command signal,
wherein the motion involves a head movement;
the head movement has an offset range within which the electric device does not respond to the head movement; and
the electric device-controlling command signal is changeable in accordance with the user's preference.

[Claim 35] A device controller comprising:
a detector configured to detect a motion of a user to generate a detection signal; and
a controller configured to process the detection signal to generate and transmit a device-controlling command signal,
wherein the motion involves a head movement;
the head movement has an offset range within which the device does not respond to the head movement; and
the device-controlling command signal is changeable in accordance with the user's preference.

[Claim 36] The device controller according to claim 35, wherein the detector is attached to eyeglasses and the head movement involves a head tilt angle;
the offset range of the head tilt angle is defined by the user and an

upper limit of the offset range is 10 degrees or less;
the device-controlling command signal is modified by multiplying, by a multiplier, a value obtained by subtracting the upper limit of the offset range from the head tilt angle; and
the multiplier is selected by the user and from 10 to 50, so that 2 to 10 degrees of the head tilt angle from the upper limit of the offset range are used to fully operate the device.

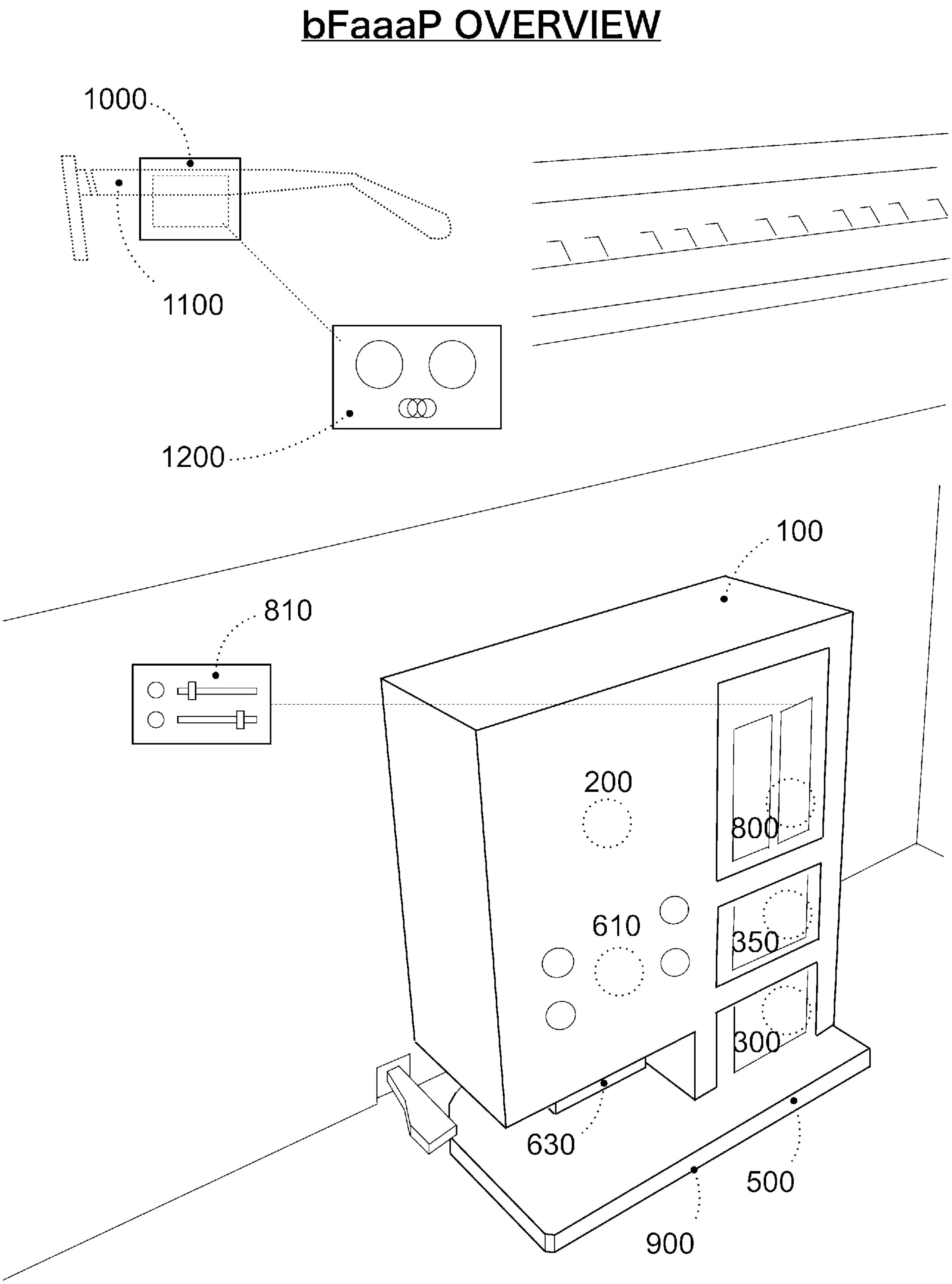
[Claim 37] The device controller according to claim 35, wherein the detector is attached to eyeglasses and the head movement involves a head tilt angle;
the offset range of the head tilt angle is defined by the user and an upper limit of the offset range is 3 degrees or less;
the device-controlling command signal is modified by multiplying, by a multiplier, a value obtained by subtracting the upper limit of the offset range from the head tilt angle; and
the multiplier is selected by the user and from 10 to 50, so that 2 to 10 degrees of the head tilt angle from the upper limit of the offset range are used to fully operate the device.

[Claim 38] The device controller according to claim 36, wherein the device controller is a gaming controller.

[Claim 39] The device controller according to claim 37, wherein the device controller is a gaming controller.

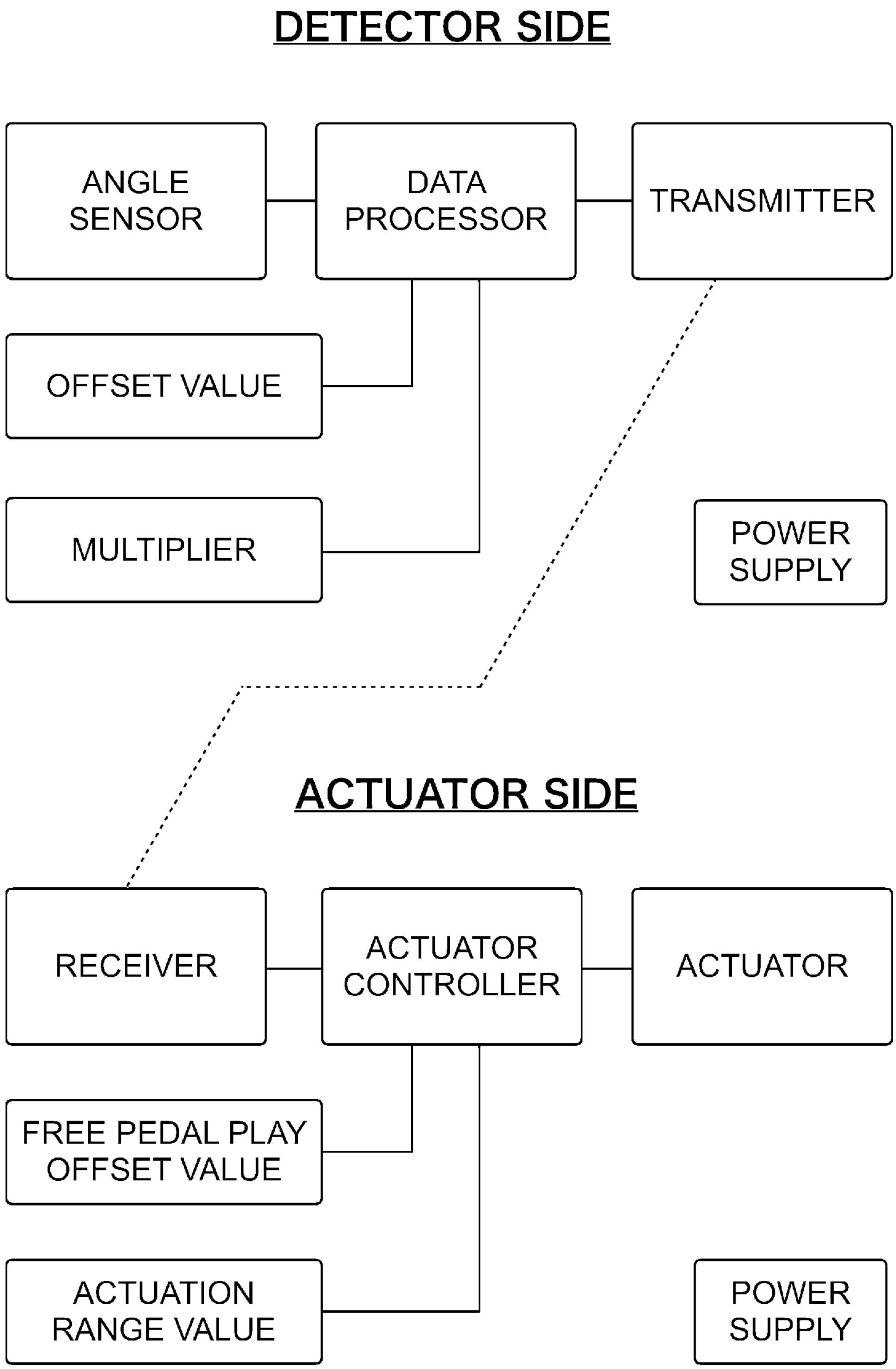
[Fig. 1]

FIG. 1


bFaaaP OVERVIEW
1000
1100
1200
100
810
200
800
610
350
300
630
500
900

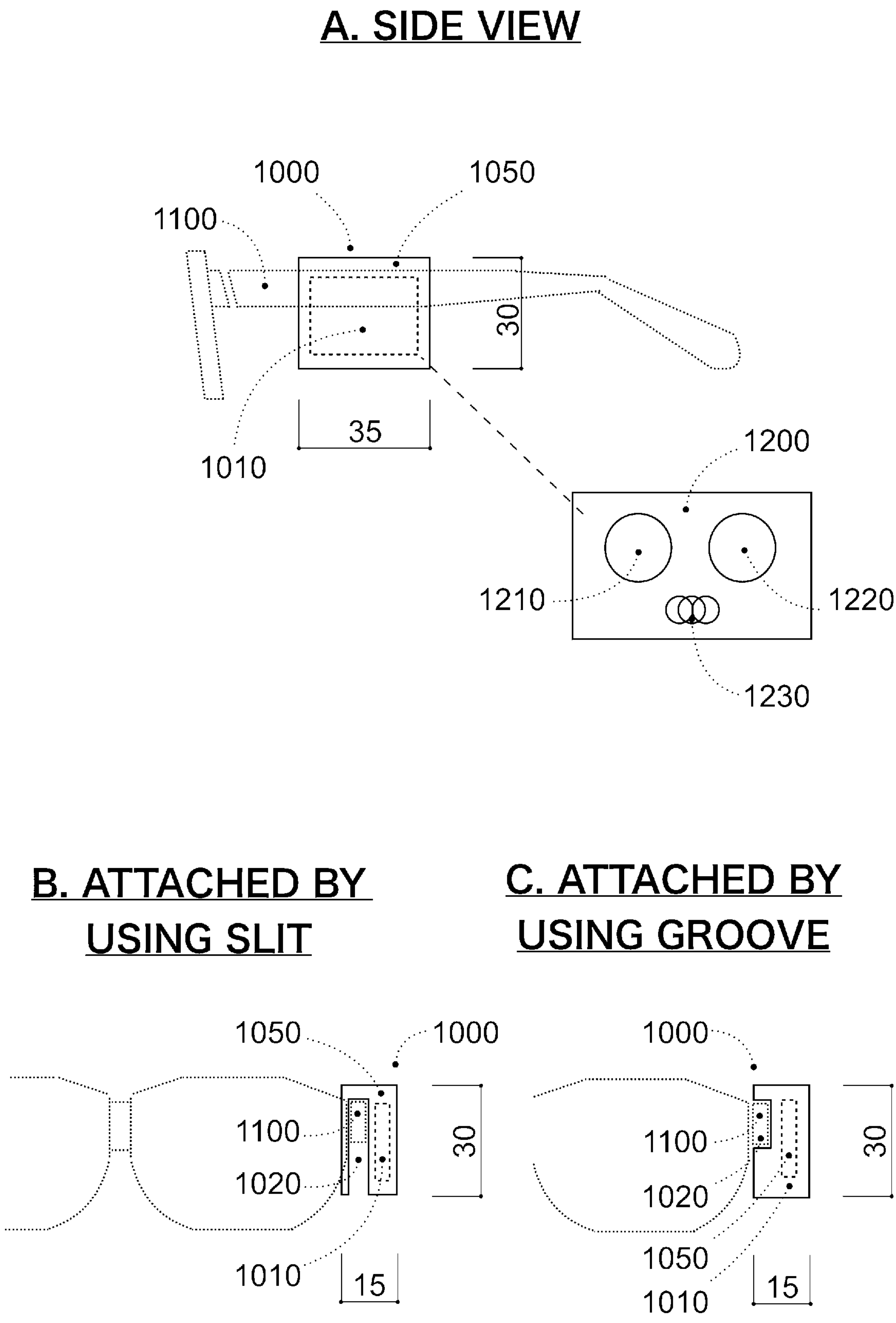
[Fig. 2]

FIG. 2


DETECTOR SIDE
ANGLE SENSOR
DATA PROCESSOR
TRANSMITTER
OFFSET VALUE
MULTIPLIER
POWER SUPPLY
ACTUATOR SIDE
RECEIVER
ACTUATOR CONTROLLER
ACTUATOR
FREE PEDAL PLAY OFFSET VALUE
ACTUATION RANGE VALUE
POWER SUPPLY

[Fig. 3]

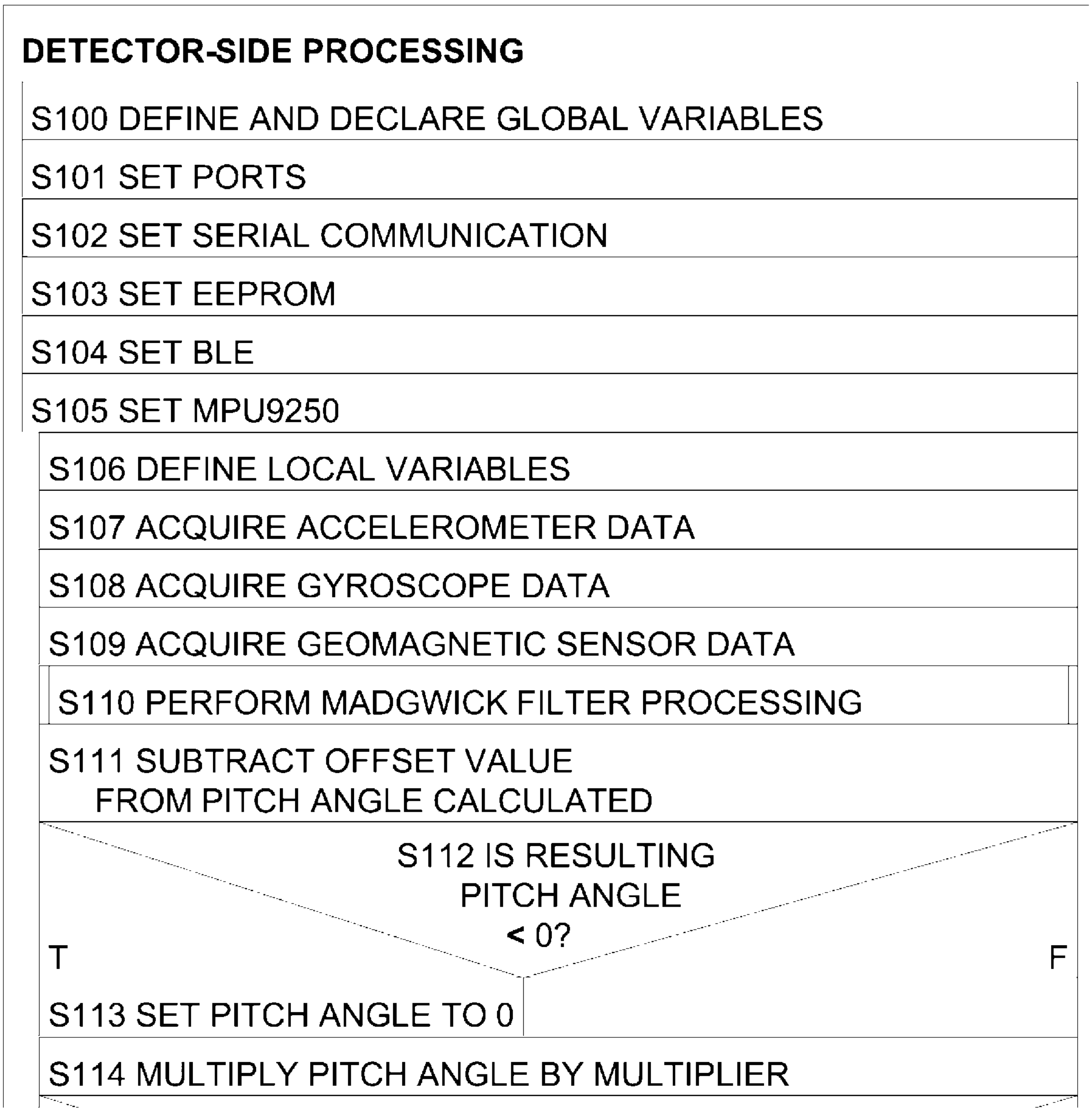
FIG. 3


A. SIDE VIEW
1000
1050
1100
30
35
1010
1200
1210
1220
1230
B. ATTACHED BY USING SLIT
1050
1000
1100
1020
1010
30
15
C. ATTACHED BY USING GROOVE
1000
1100
1020
1050
1010
30
15

[Fig. 4]

FIG. 4

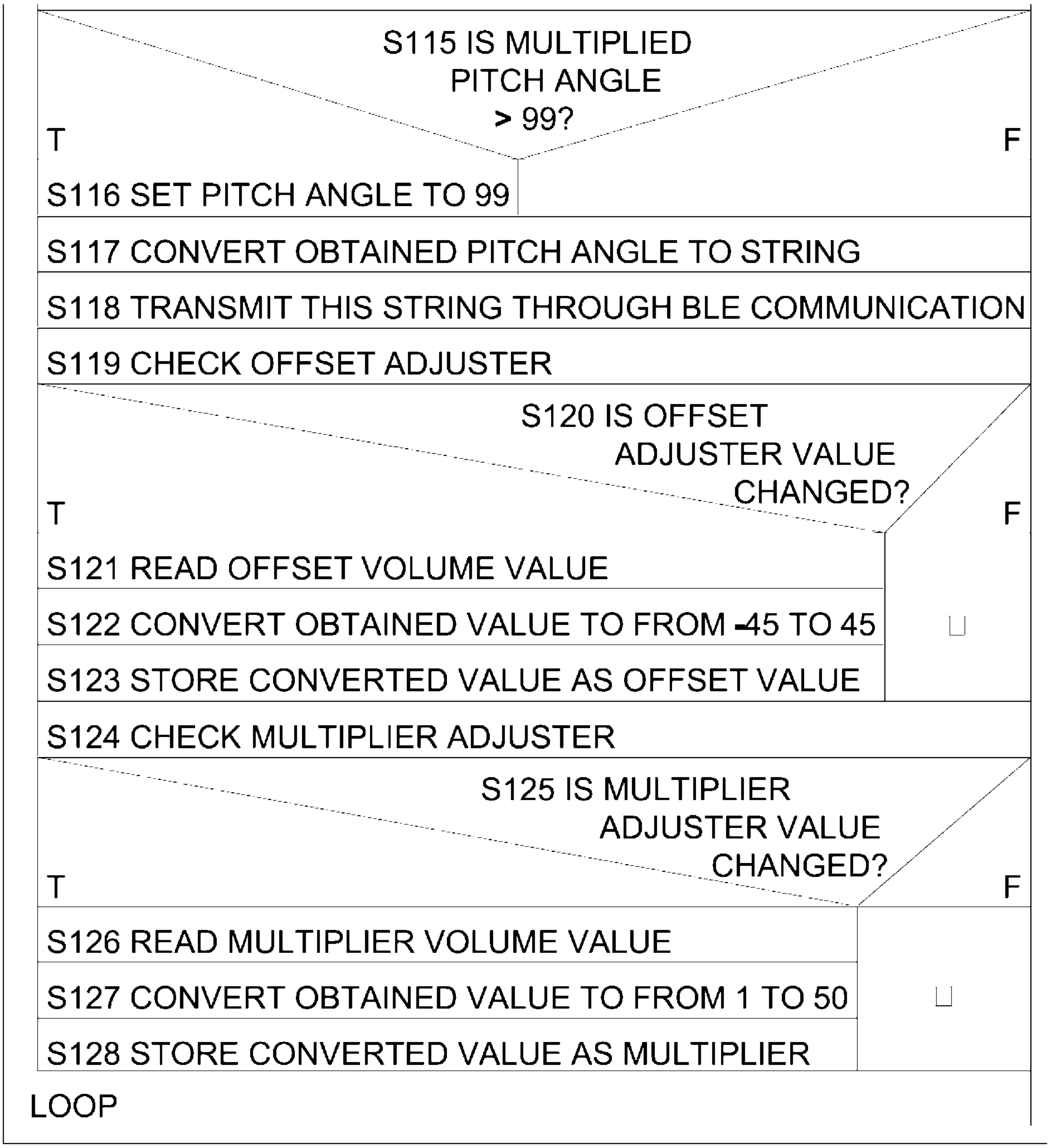
DETECTOR-SIDE PROCESSING FLOWCHART I



[Fig. 5]

FIG. 5

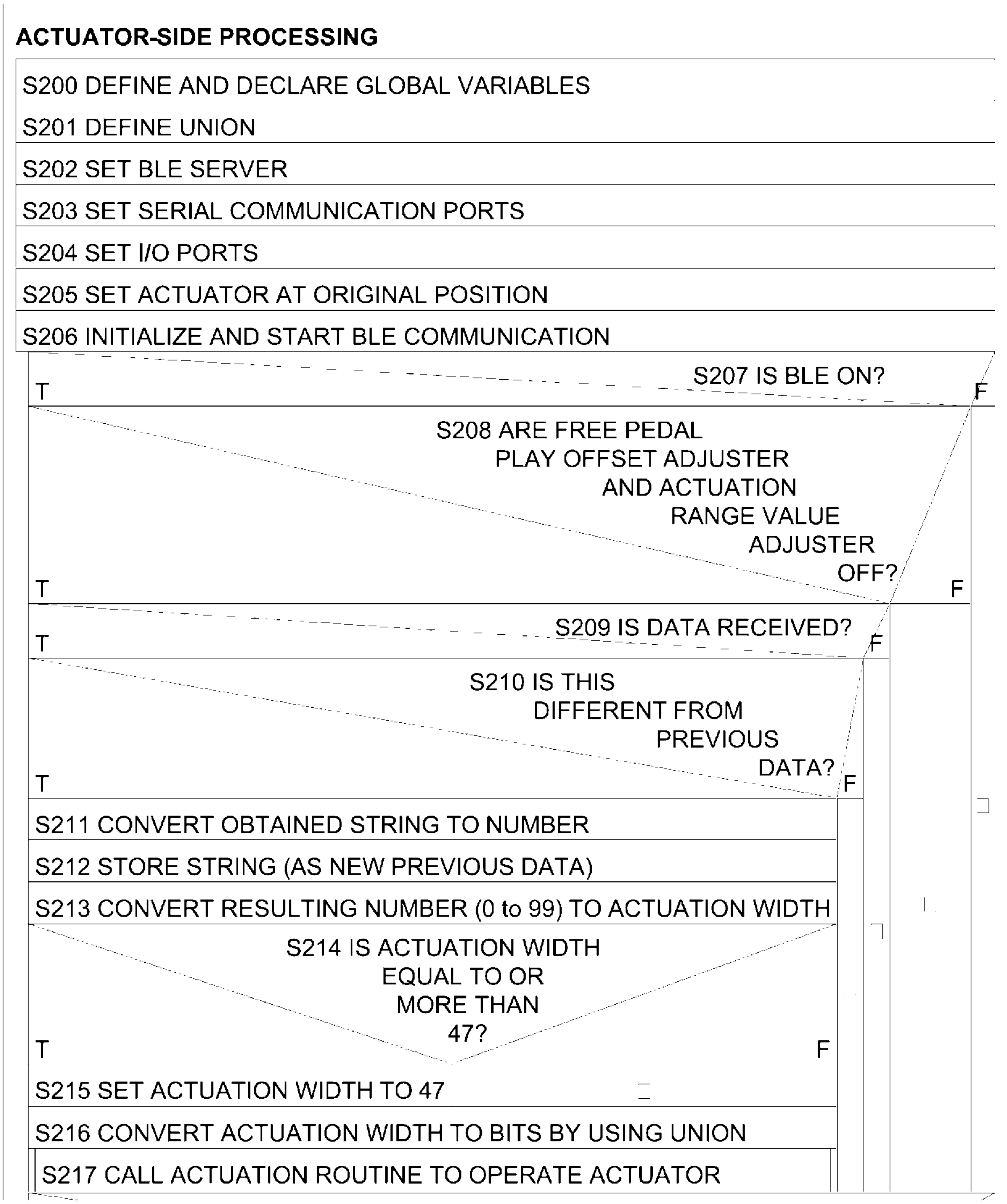
DETECTOR-SIDE PROCESSING FLOWCHART II



[Fig. 6]

FIG. 6

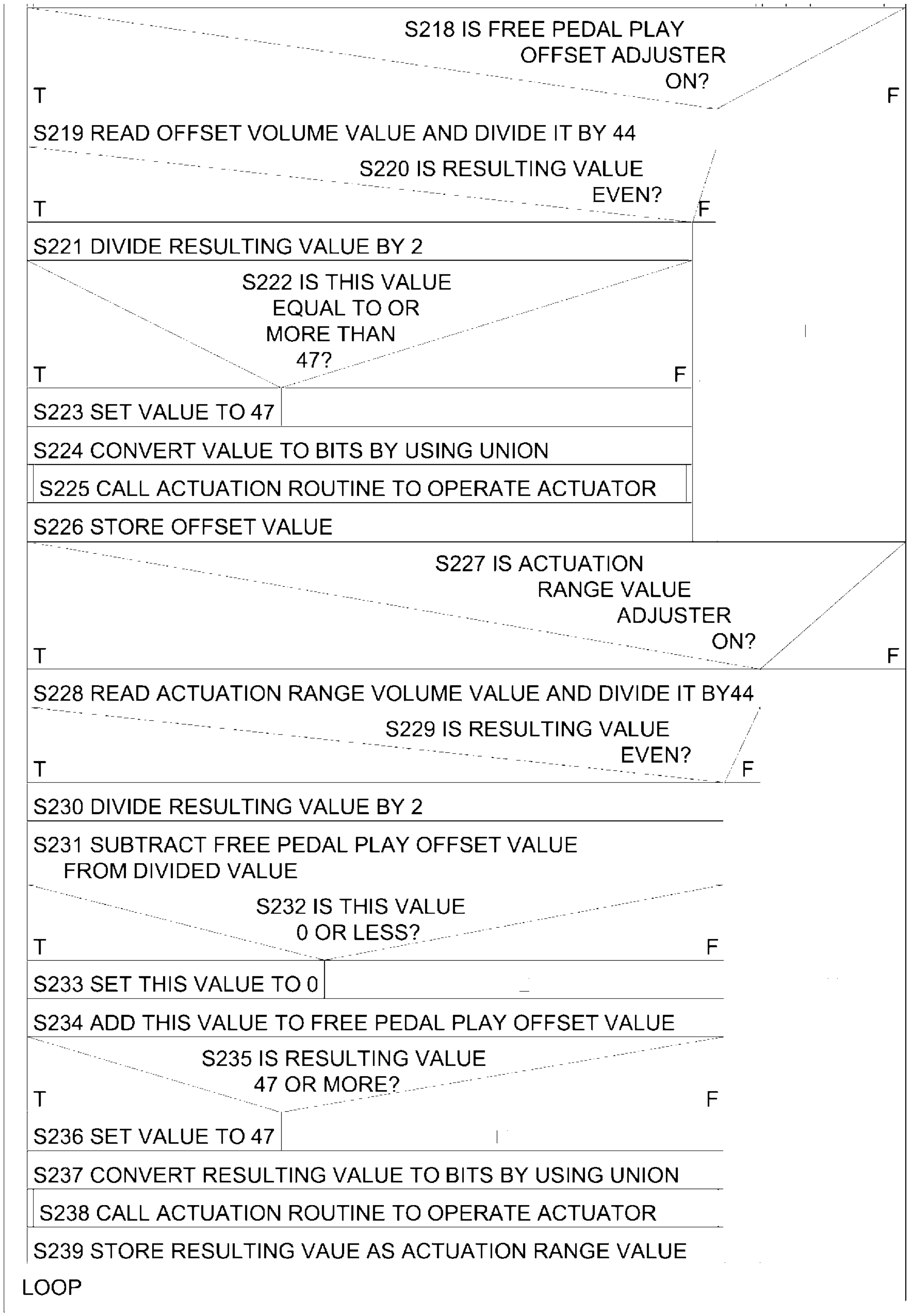
ACTUATOR-SIDE PROCESSING FLOWCHART I



[Fig. 7]

FIG. 7

ACTUATOR-SIDE PROCESSING FLOWCHART II



[Fig. 8]

FIG. 8

MOV FUNCTION FLOWCHART

MOV FUNCTION PROCESSING

S300 WHILE ACTUATOR IS BUSY

S301 SET Mo0 BIT

S302 SET Mo1 BIT

S303 SET Mo2 BIT

S304 SET Mo3 BIT

S305 SET Mo4 BIT

S306 SET Mo5 BIT

S307 DELAY 5 ms

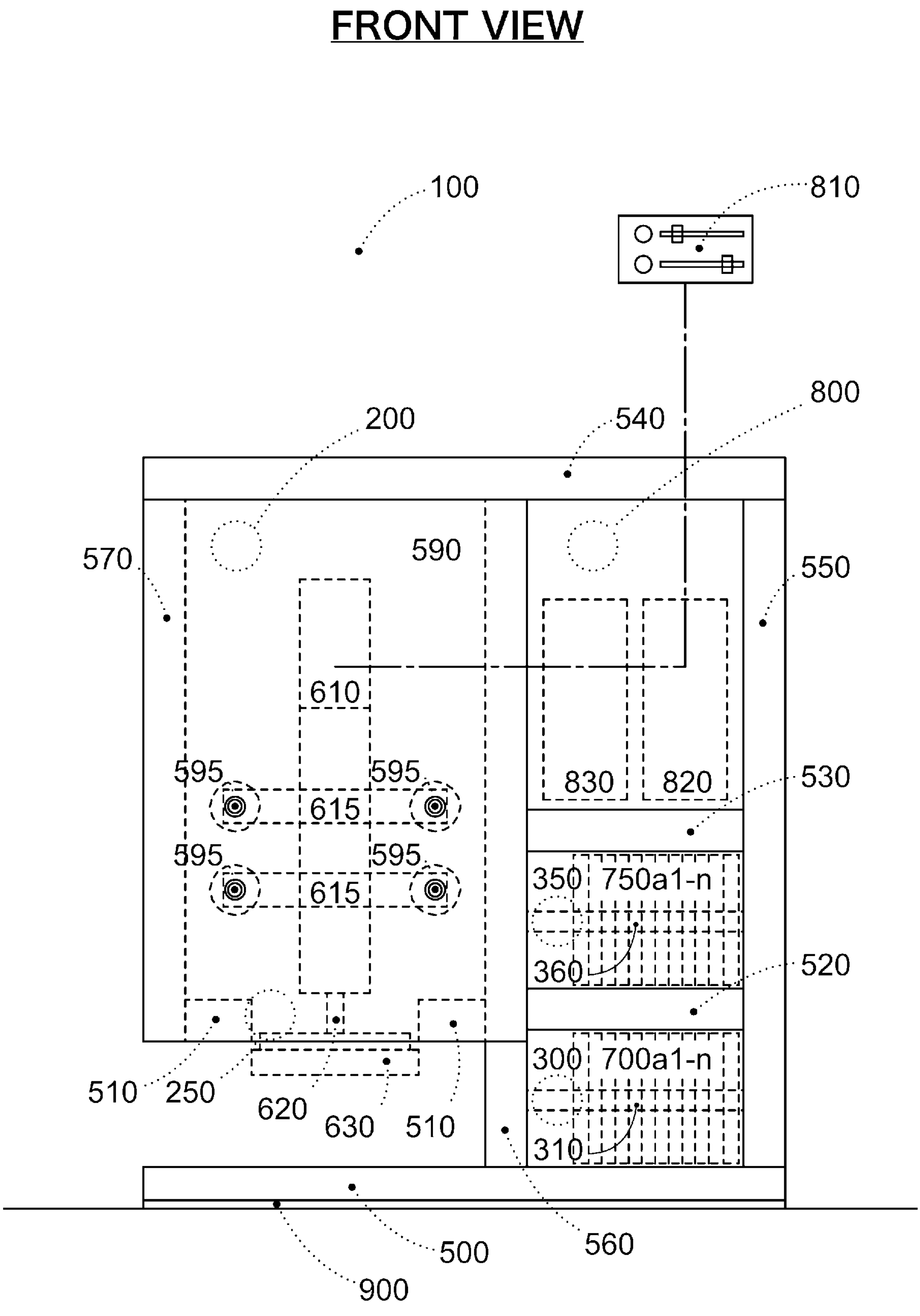
S308 TURN START BIT ON

S309 WHILE ACTUATOR IS BUSY

S310 TURN START BIT OFF

[Fig. 9]

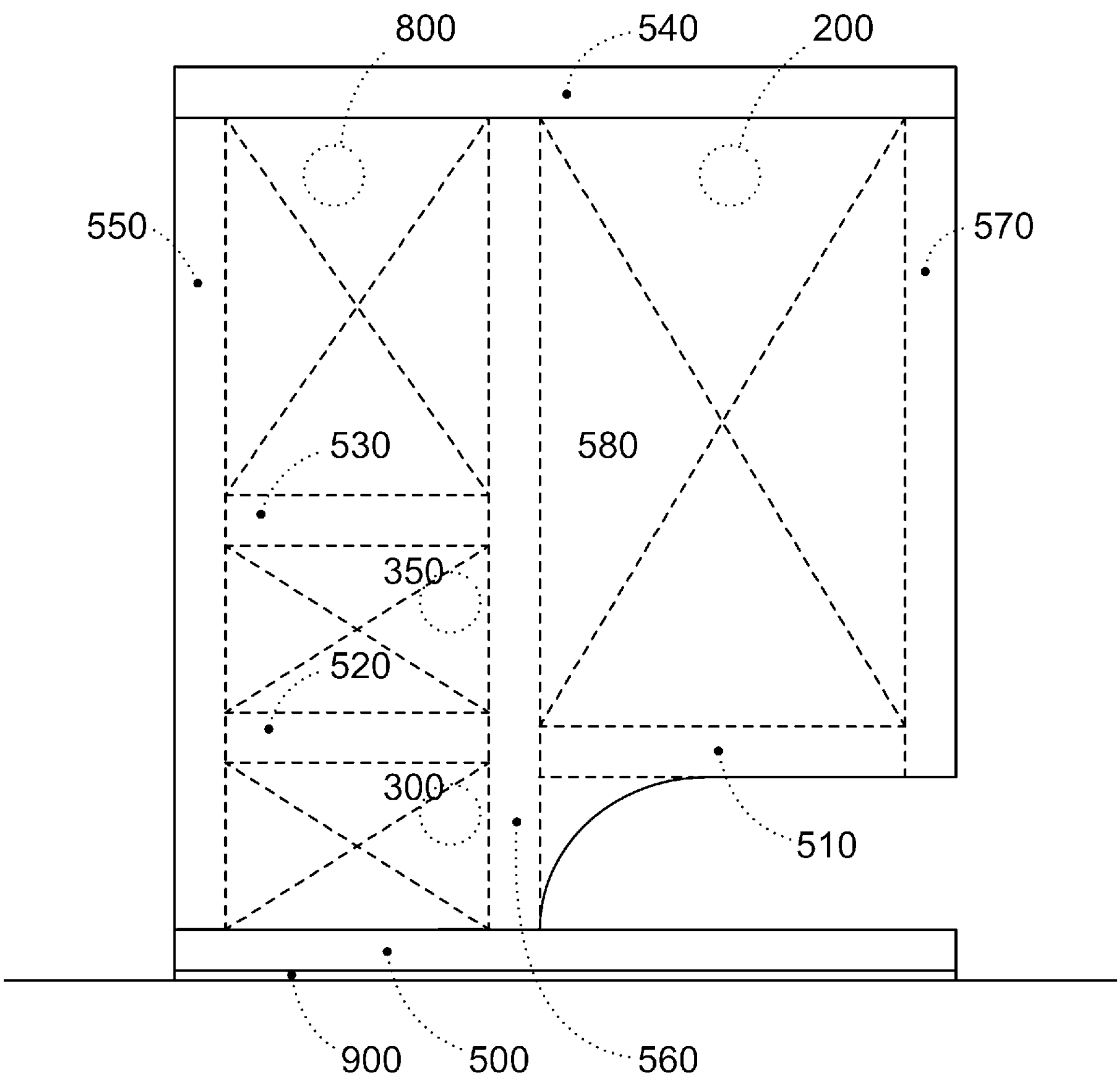
FIG. 9


FRONT VIEW
100
810
800
200
540
570
590
550
610
830
820
595
595
615
530
595
595
350
750a1-n
615
360
520
300
700a1-n
510
250
620
630
510
310
500
560
900

[Fig. 10]

FIG. 10

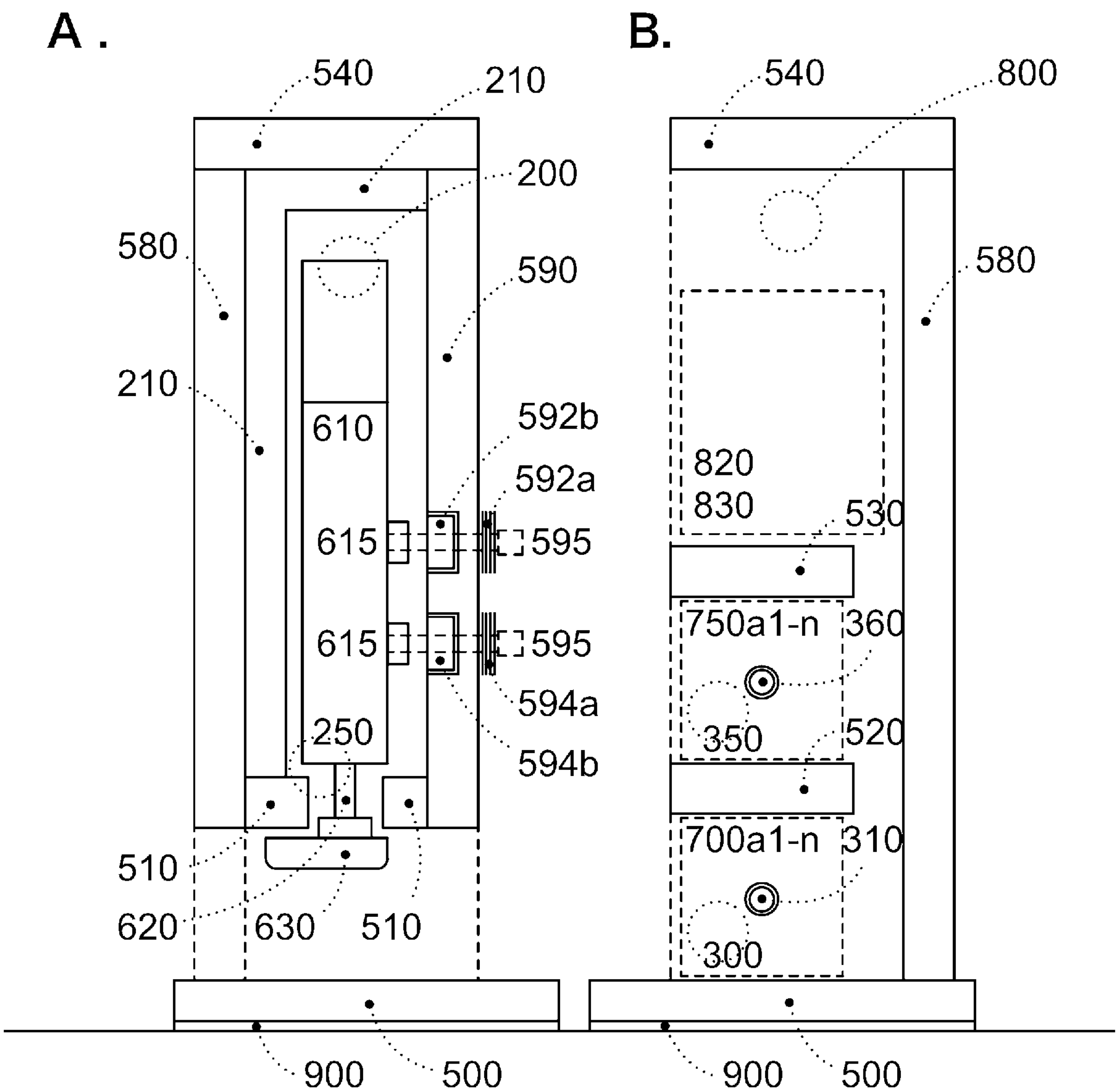
REAR VIEW



[Fig. 11]

FIG. 11

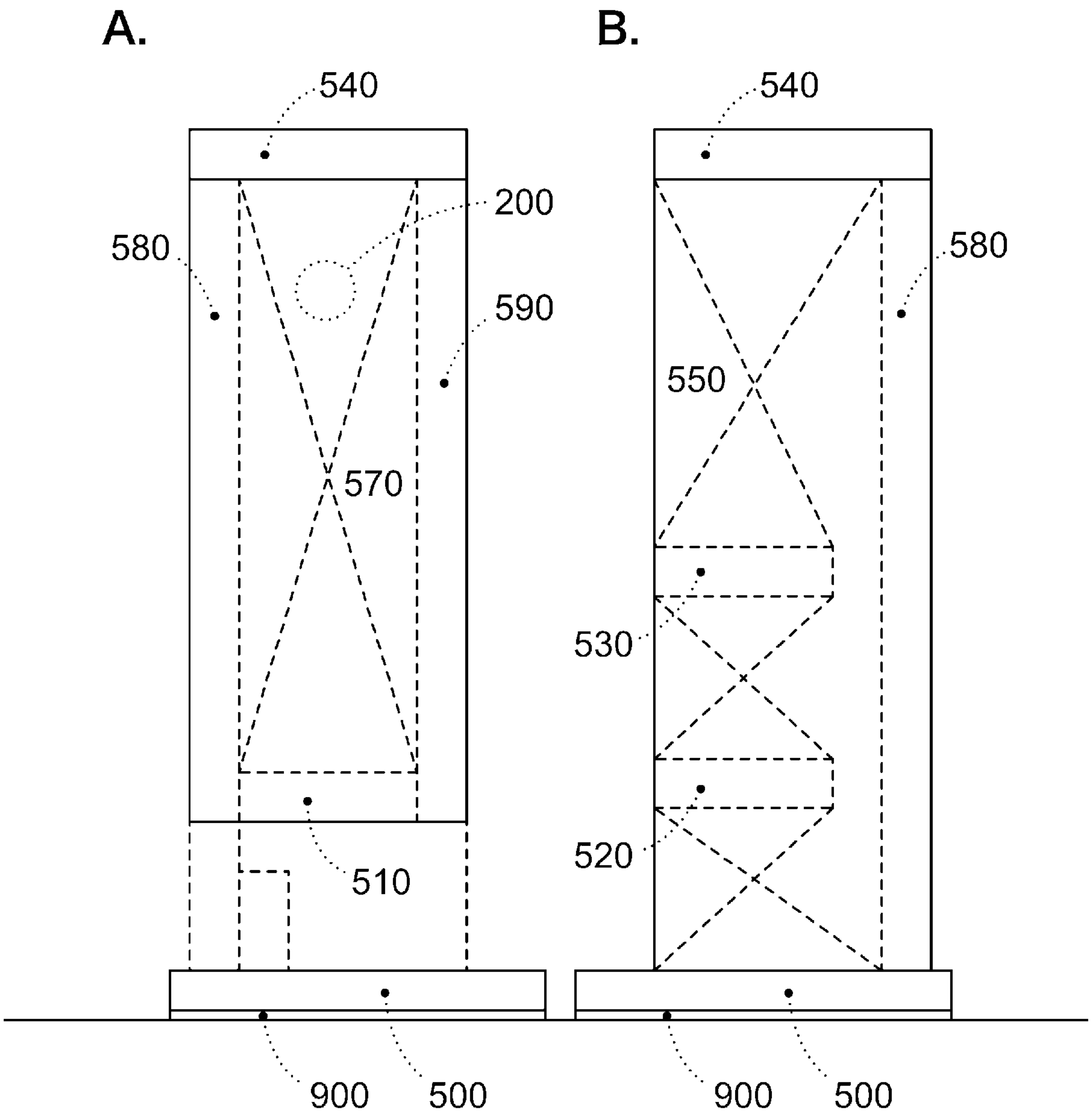
LEFT-SIDE AND RIGHT-SIDE CROSS-SECTIONAL VIEWS



[Fig. 12]

FIG. 12

LEFT-SIDE AND RIGHT-SIDE VIEWS

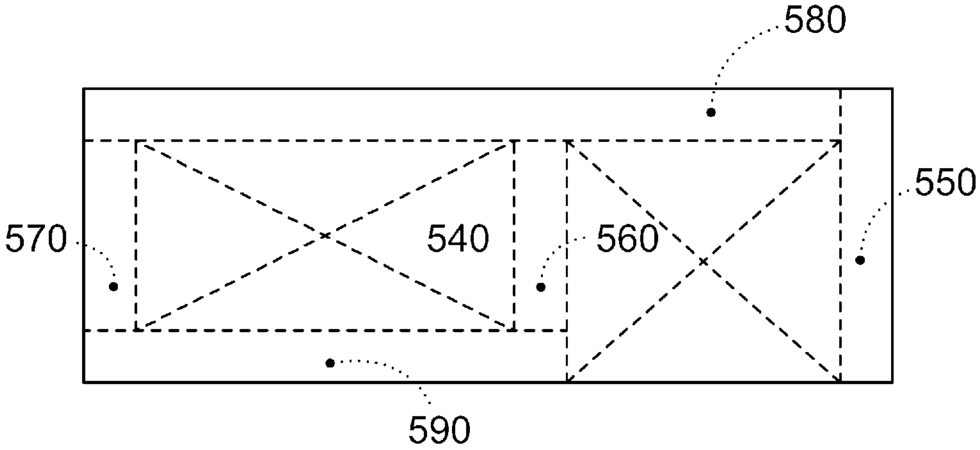


[Fig. 13]

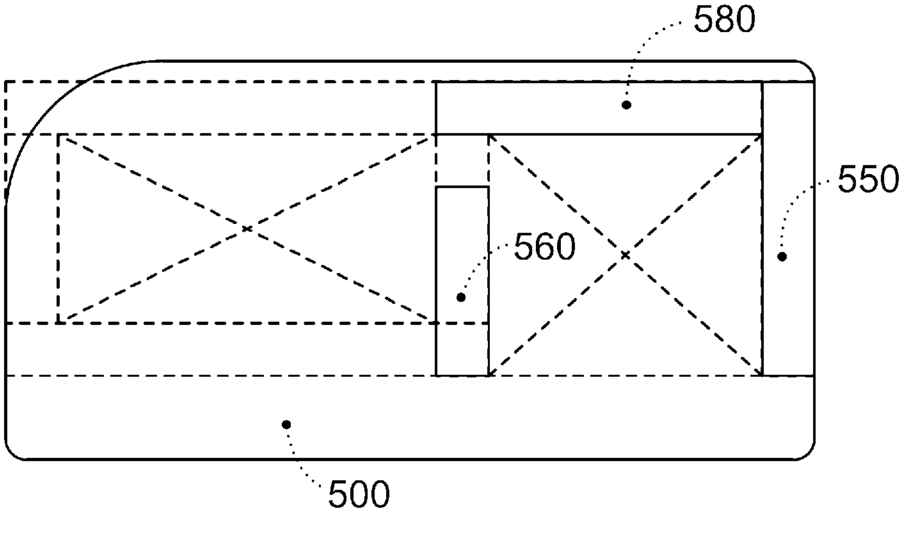
FIG. 13

CROSS-SECTIONAL VIEWS

A. TOP VIEW



B. BOTTOM VIEW

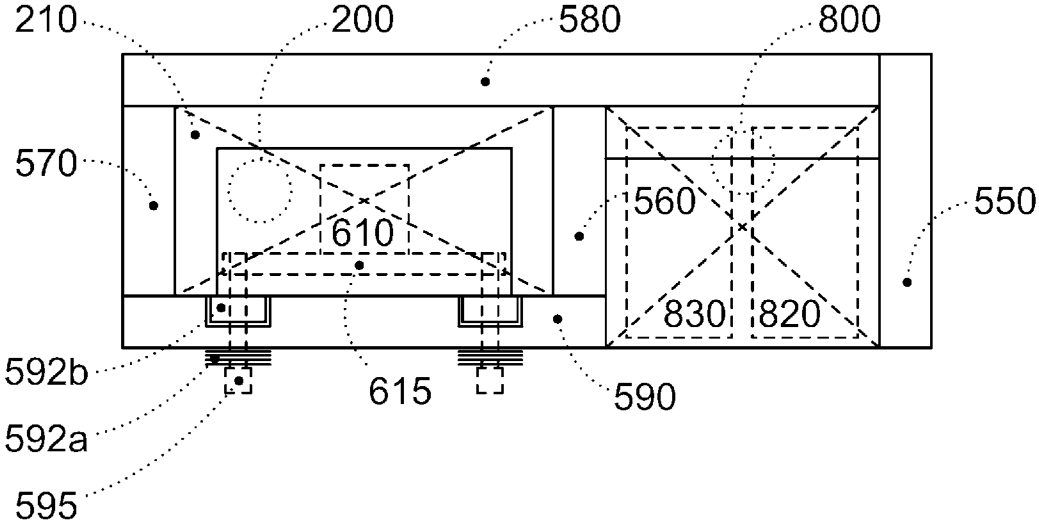


[Fig. 14]

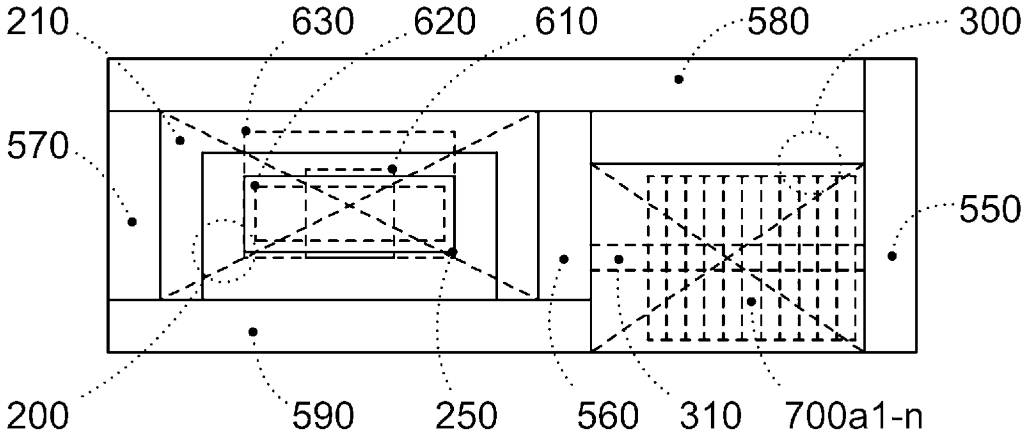
FIG.14

CROSS-SECTIONAL VIEWS

A. INTERMEDIATE VIEW I



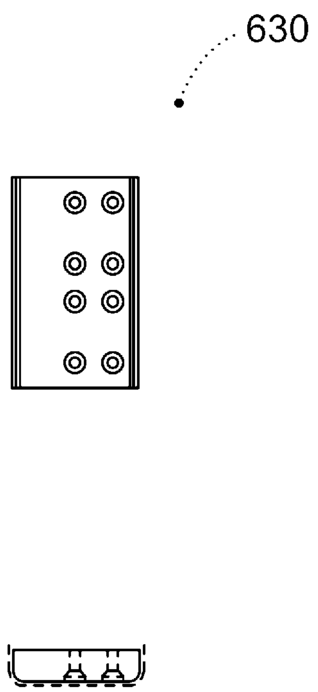
B. INTERMEDIATE VIEW II



[Fig. 15]

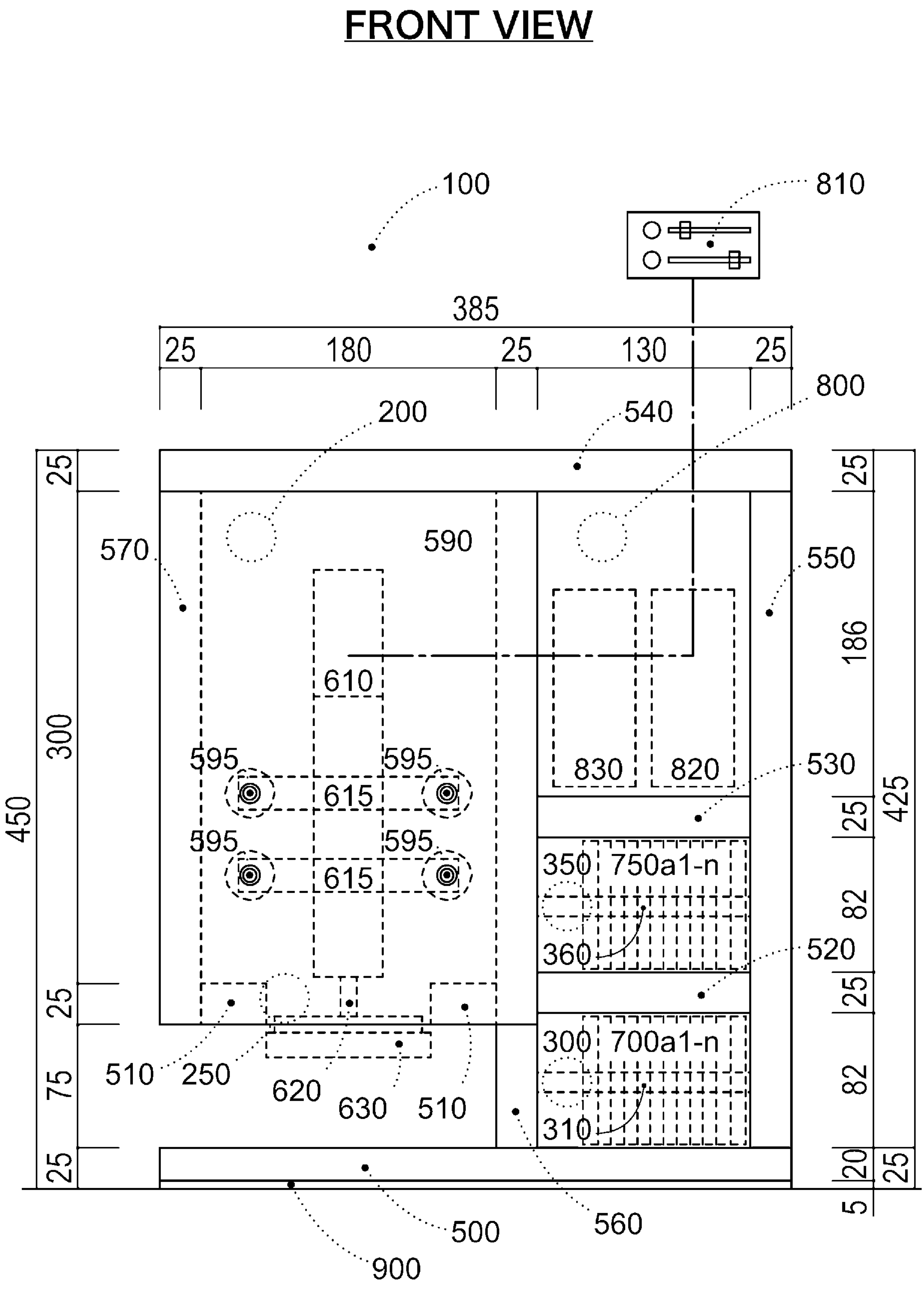
FIG. 15

ENLARGED VIEWS



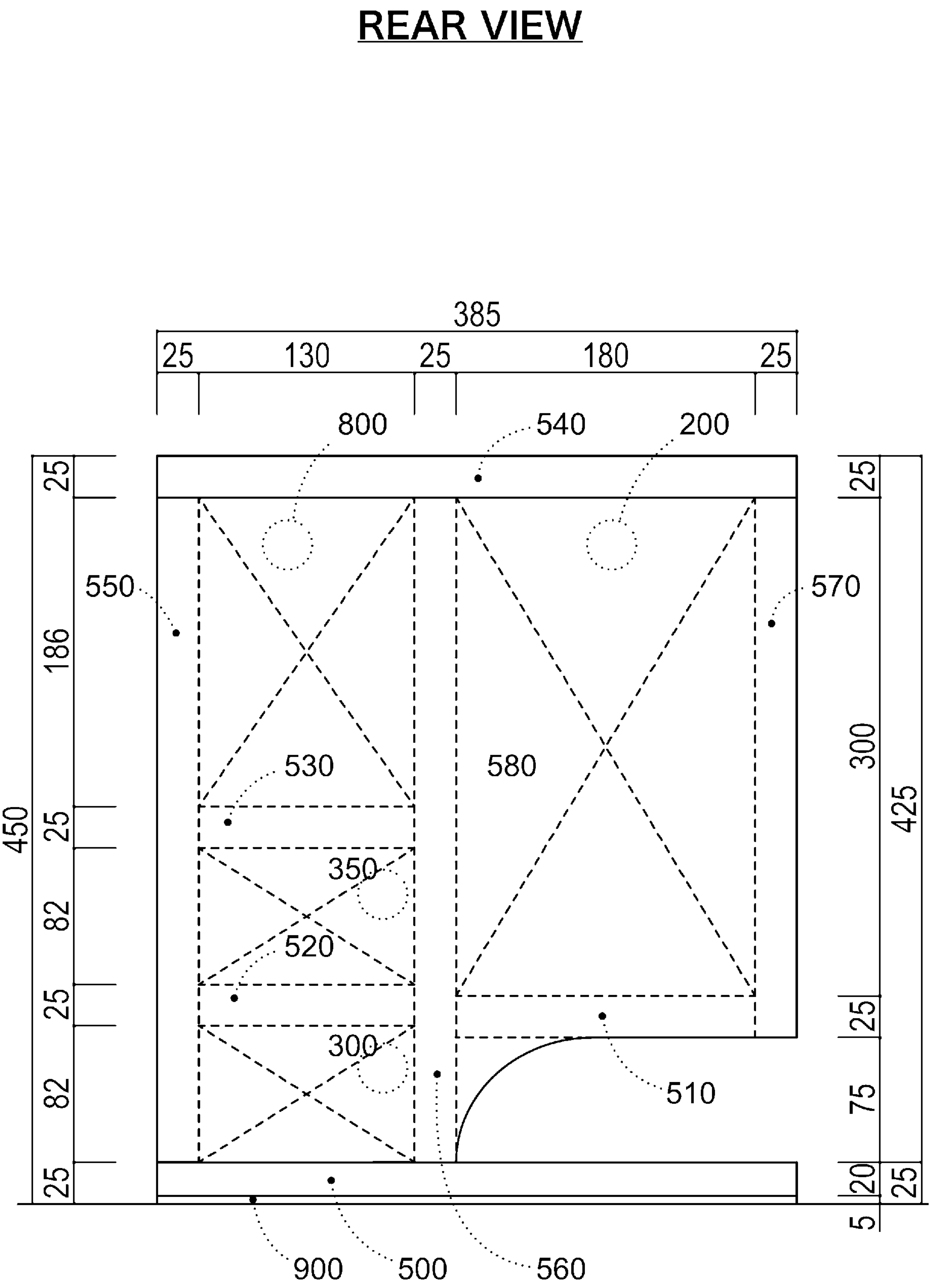
[Fig. 16]

FIG. 16


FRONT VIEW
100
810
385
25
180
25
130
25
200
540
800
570
590
550
610
186
300
450
425
595
595
615
830
820
530
595
595
615
350
750a1-n
360
82
520
25
25
300
700a1-n
510
250
620
630
510
310
75
82
20
25
5
500
560
900

[Fig. 17]

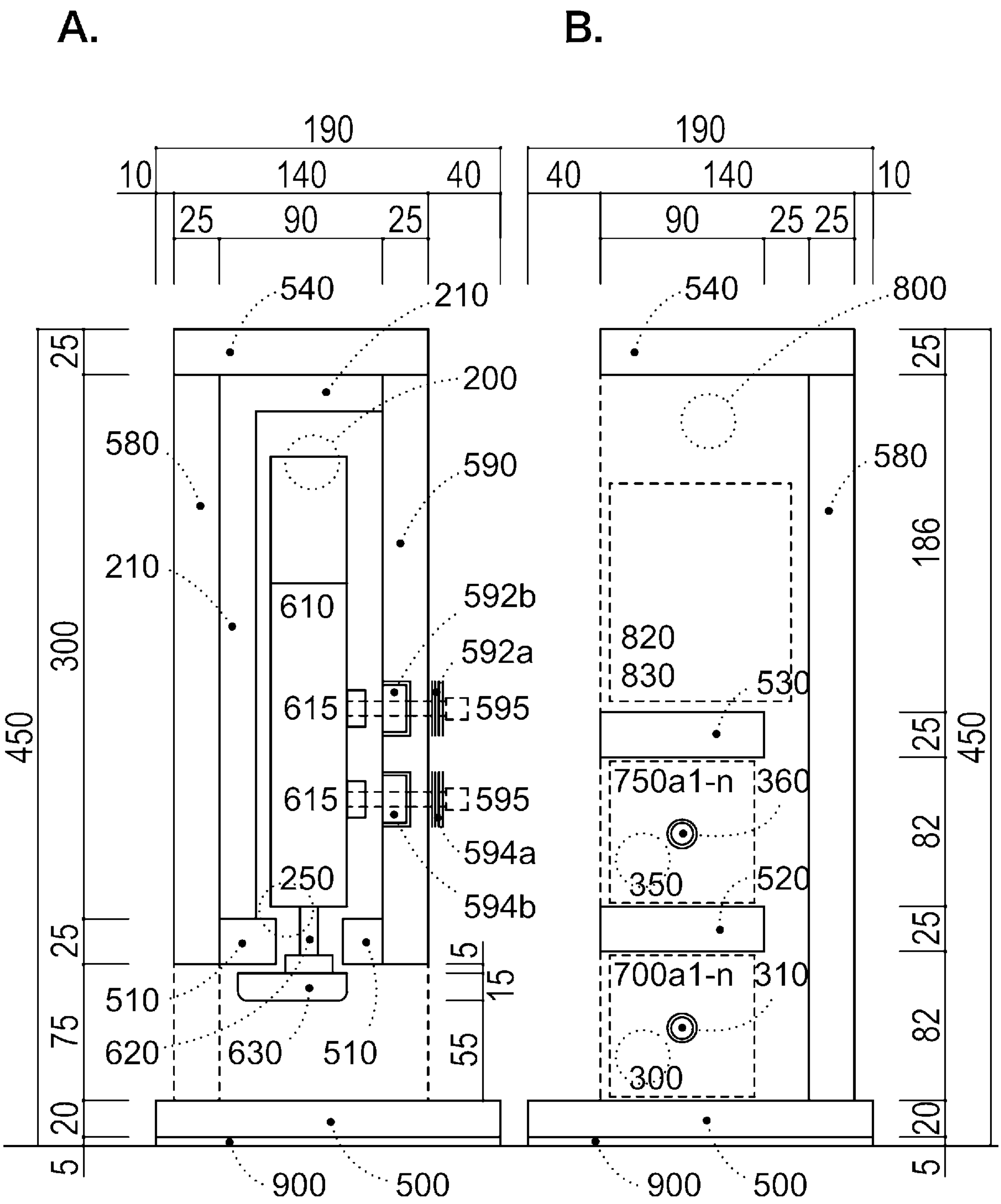
FIG. 17


REAR VIEW
385
25
130
25
180
25
800
540
200
25
25
550
570
186
530
580
300
425
450
25
350
82
520
25
25
300
510
82
75
25
20
25
5
900
500
560

[Fig. 18]

FIG. 18

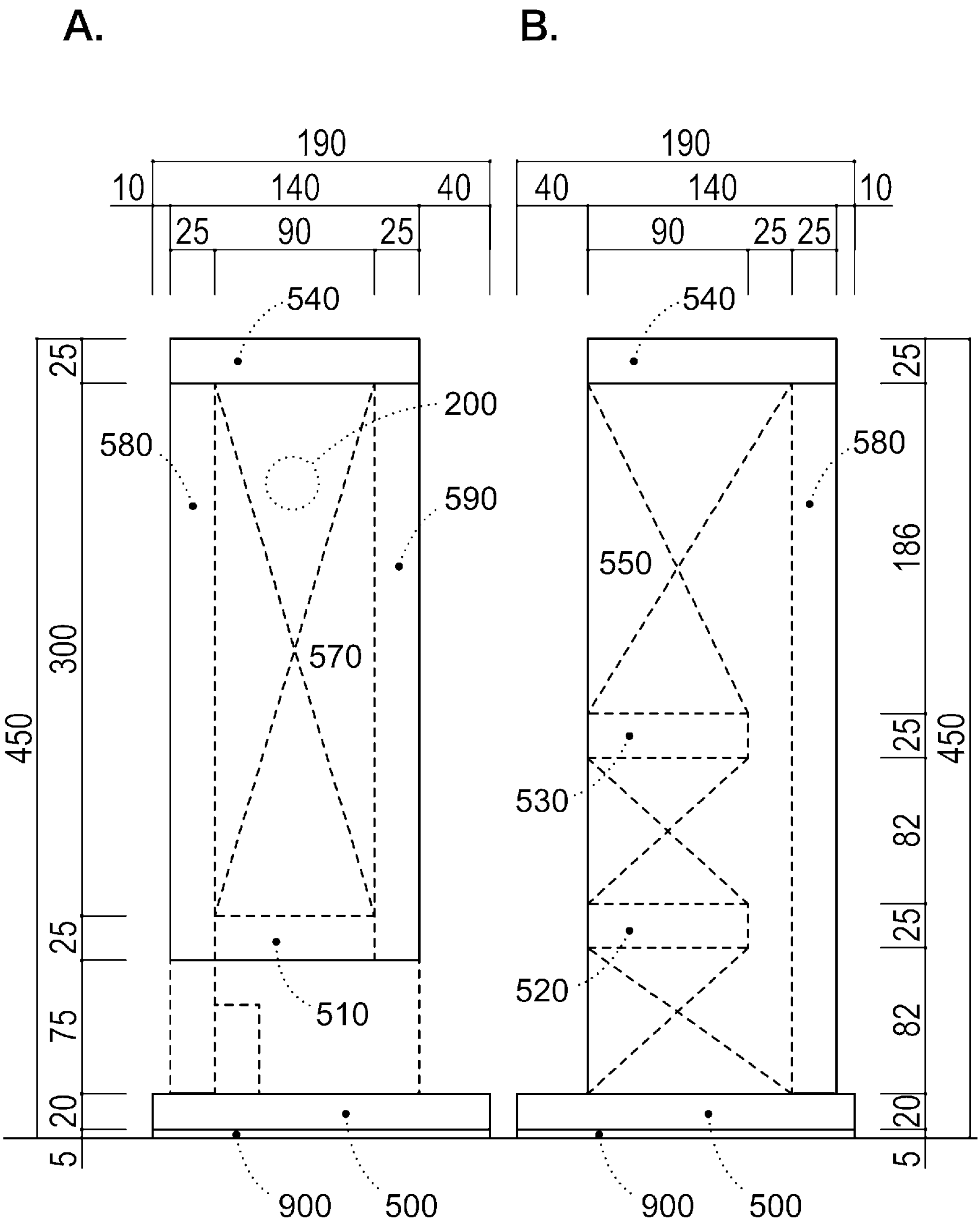
LEFT-SIDE AND RIGHT-SIDE CROSS-SECTIONAL VIEWS



[Fig. 19]

FIG. 19

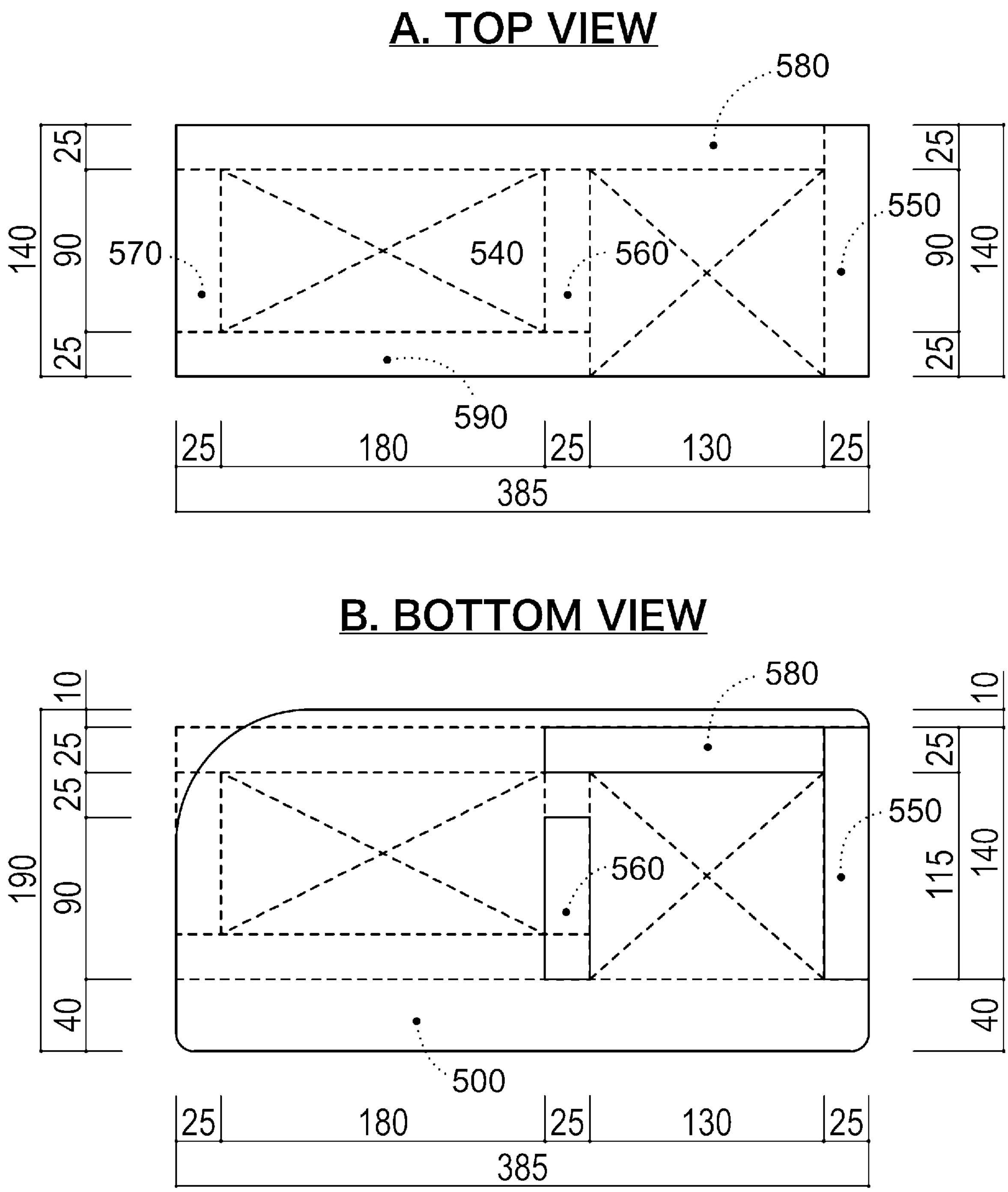
LEFT-SIDE AND RIGHT-SIDE VIEWS



[Fig. 20]

FIG. 20

CROSS-SECTIONAL VIEWS

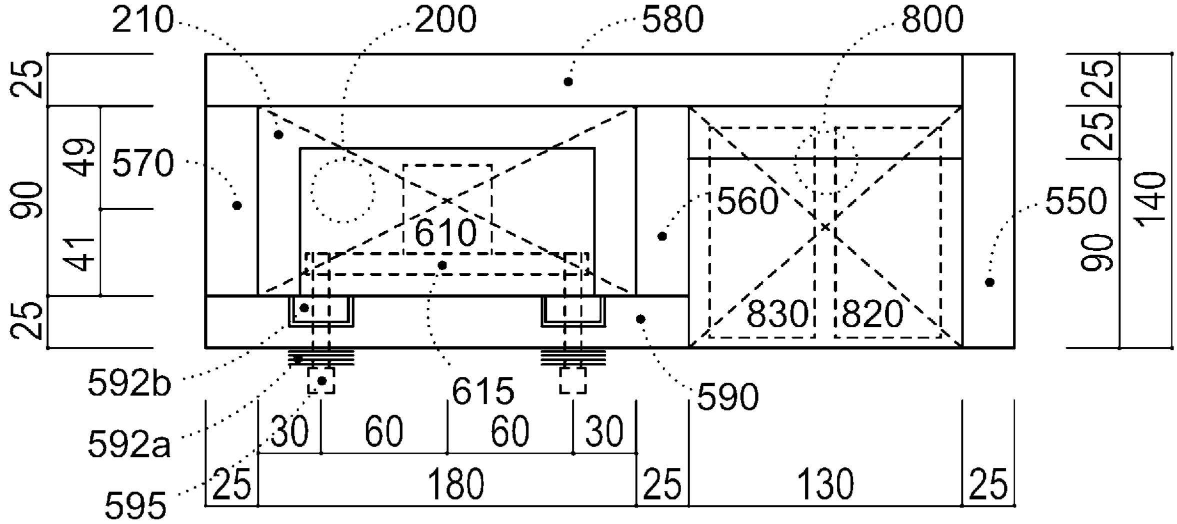


[Fig. 21]

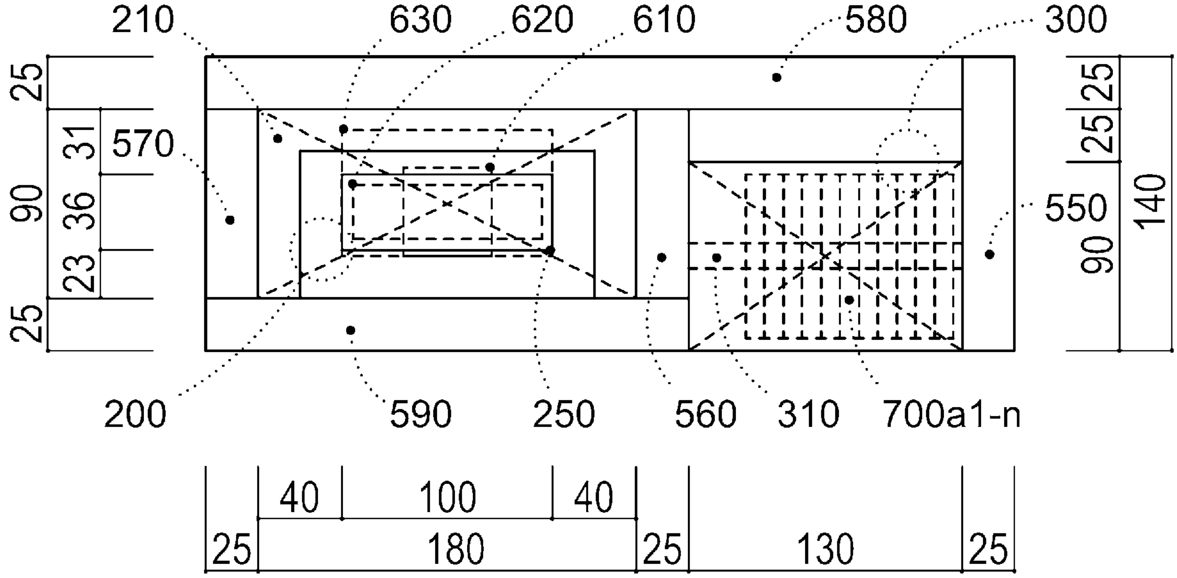
FIG. 21

CROSS-SECTIONAL VIEWS

A. INTERMEDIATE VIEW I



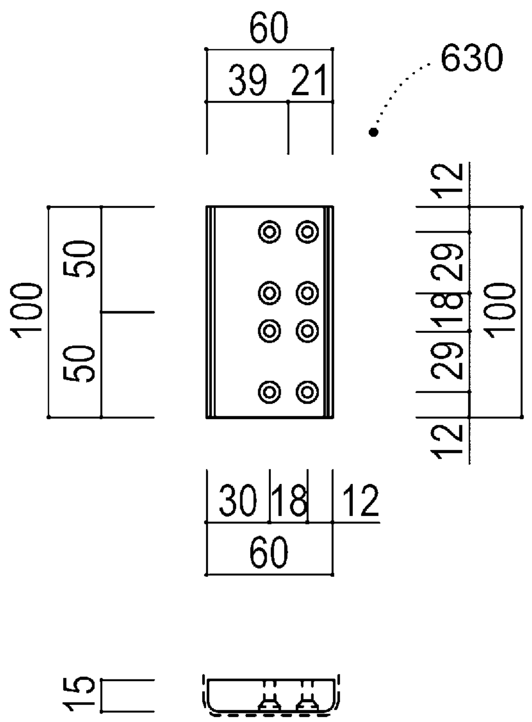
B. INTERMEDIATE VIEW II



[Fig. 22]

FIG. 22

ENLARGED VIEWS



[Fig. 23]

FIG. 23

NOTES PLAYED IN APEE STUDY

[Fig. 24]

FIG. 24

QUESTIONNAIRE IN PROTOCOL OF APEE STUDY

<table>
<tr><td colspan="2">Your Name：</td><td colspan="2">Age：</td><td colspan="2">Sex: Man・Woman</td></tr>
<tr><td colspan="3">Test Date：</td><td colspan="3">Piano Experience：　　Years</td></tr>
<tr><td colspan="6">Degree of Disabilities： Wheelchair・Others</td></tr>
<tr><td colspan="6">Offset value (OFFSET)：　　Multiplier (MULT)：</td></tr>
<tr><td colspan="3">Impression</td><td>Good</td><td>Intermediate</td><td>Poor</td></tr>
<tr><td>1</td><td colspan="2">Did you enjoy it？</td><td>A</td><td>B</td><td>C</td></tr>
<tr><td>2</td><td colspan="2">Easy to use？</td><td>A</td><td>B</td><td>C</td></tr>
<tr><td>3</td><td colspan="2">Good appearance？</td><td>A</td><td>B</td><td>C</td></tr>
<tr><td>4</td><td colspan="2">Does the name of bFaaaP sound good？</td><td>A</td><td>B</td><td>C</td></tr>
<tr><td colspan="6">If you have any other comments, please provide them below.</td></tr>
</table>

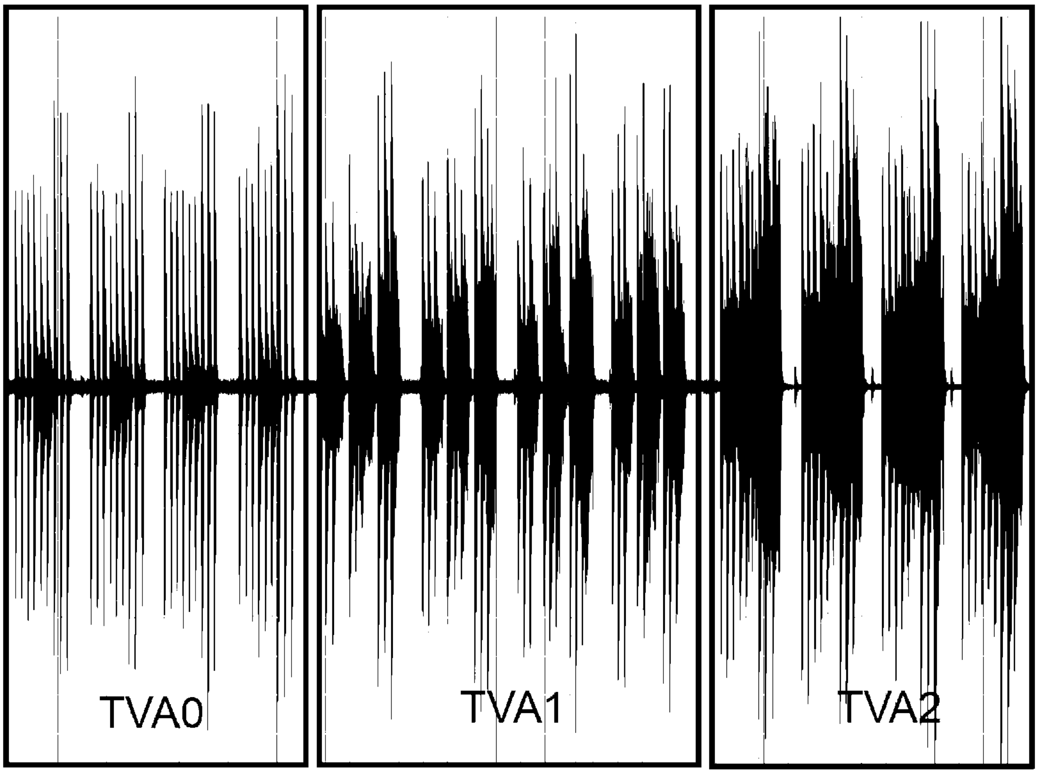
I agree that the content and results of this test are not disclosed publicly until the patent filing.

[Fig. 25]

FIG. 25

APEE STUDY RESULTS

A.

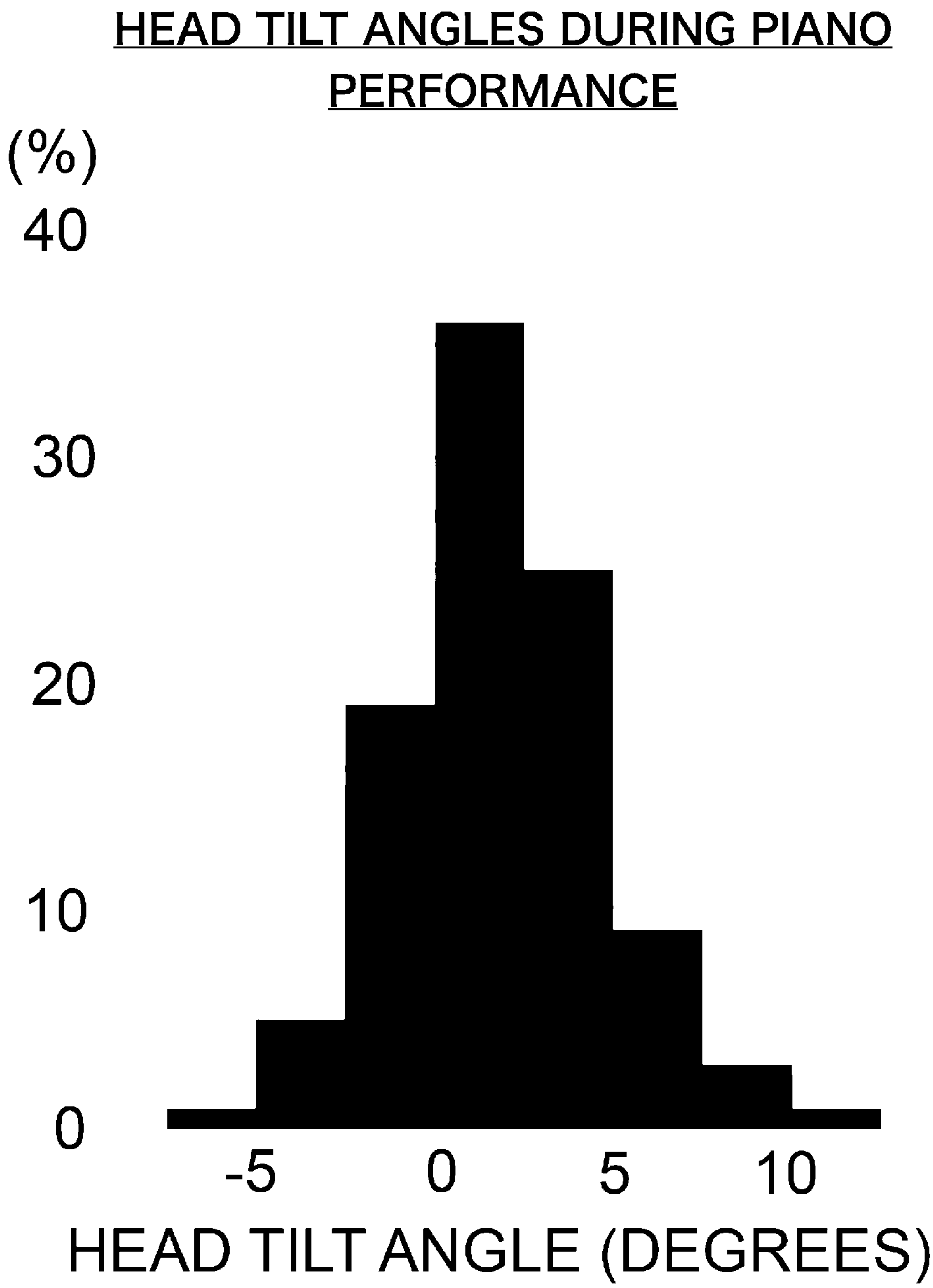


B.

TVA0 IS SET TO 1
SCORE (PEDAL PATTERN 1) = TVA1 / TVA0
SCORE (PEDAL PATTERN 2) = TVA2 / TVA0

[Fig. 26]

FIG. 26


HEAD TILT ANGLES DURING PIANO PERFORMANCE
(%)
40
30
20
10
0
-5
0
5
10
HEAD TILT ANGLE (DEGREES)

INTERNATIONAL SEARCH REPORT

International application No.
PCT/JP2018/041771

A. CLASSIFICATION OF SUBJECT MATTER

Int.Cl. G06F3/01(2006.01)i, A63F13/24(2014.01)i, G10H1/00(2006.01)i

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

Int.Cl. G06F3/01, A63F13/24, G10H1/00

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Published examined utility model applications of Japan 1922-1996
Published unexamined utility model applications of Japan 1971-2019
Registered utility model specifications of Japan 1996-2019
Published registered utility model applications of Japan 1994-2019

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
Y	Head-Activated Piano Pedal, [online], 2018.04.02 [retrieved on 2019.01.10], Retrieved from the Internet: <URL: https://web.archive.org/web/20180402202115/https://www.canassist.ca/EN/main/programs/technologies-and-devices/test-1/piano-arts.html>, 4th paragraph	1-39
Y	JP 2017-37567 A (COLOPL, INC.) 2017.02.16, paragraphs [0012], [0018], [0021]-[0023] (Family: none)	1-39
Y	JP 2014-95953 A (TOKYO INSTITUTE OF TECHNOLOGY) 2014.05.22, paragraphs [0029], [0043] & WO 2014/073121 A1	1-39

☑ Further documents are listed in the continuation of Box C. ☐ See patent family annex.

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"O" document referring to an oral disclosure, use, exhibition or other means
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"&" document member of the same patent family

Date of the actual completion of the international search	Date of mailing of the international search report
16.01.2019	29.01.2019

Name and mailing address of the ISA/JP	Authorized officer
Japan Patent Office	SAKURAI, Shigeyuki — 5E 7887
3-4-3, Kasumigaseki, Chiyoda-ku, Tokyo 100-8915, Japan	Telephone No. +81-3-3581-1101 Ext. 3521

C (Continuation). DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
Y	US 2018/0064371 A1 (ALPS ELECTRIC CO., LTD.) 2018.03.08, paragraphs [0030], [0034]-[0035], FIG. 1 & WO 2016/194581 A1 & EP 3305193 A1 & CN 107708552 A	2-3, 8-11,14-15, 18-19, 22-23, 26-27, 30-33, 36-39
Y	JP 2006-154504 A (YAMAHA CORPORATION) 2006.06.15, paragraphs [0015], [0018], [0027], [0047]-[0049], [0063], FIG. 9(a) & US 2006/0112809 A1, paragraphs [0077]-[0079], [0102], [0115]-[0118], [0122], FIG. 8C	20-27
A	US 4736664 A (Hinsley; J. D., Norwood; Bobby) 1988.04.12, Whole Document (Family: none)	1-39